

Economics 240A: Econometrics

Lecture Notes, Part I*

Marcelo Gallardo

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1 Lecture 1: Probability Theory

1.1 Sample spaces, events, and σ -algebras

Definition 1.1 (Sample space, event). The set S of all possible outcomes of an experiment is the **sample space**. An **event** is any subset $A \subset S$; it **occurs** if the realized outcome lies in A .

A sample space is **countable** if its elements biject with a subset of the integers: $S = \{H, T\}$ for a coin toss is finite, $S = (0, \infty)$ for a reaction time is uncountable. We use the usual set operations $A \cup B$, $A \cap B$, A^c , and $A \setminus B = A \cap B^c$, and take as known that they are commutative, associative, and distributive, and that DeMorgan's laws $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$ hold for arbitrary families as well as for pairs. Two decompositions recur throughout; in each the sets on the right are disjoint:

$$B = (B \cap A) \cup (B \cap A^c), \quad (1)$$

$$A \cup B = A \cup (B \cap A^c). \quad (2)$$

The first drives the law of total probability (Theorem 1.12); the second turns a union into a disjoint union so that countable additivity applies.

Definition 1.2 (Disjointness, partition). Events are **pairwise disjoint** if $A_i \cap A_j = \emptyset$ for $i \neq j$; pairwise disjoint events with $\bigcup_{i=1}^{\infty} A_i = S$ form a **partition** of S .

We would like $\mathbb{P}$ to be defined on *every* subset of S . When S is countable this is harmless and we simply take $\mathcal{B} = 2^S$; when S is uncountable it turns out to be impossible. Take $S = [0, 1]$ and ask for the two properties any sensible notion of “length” must have:

$$\mathbb{P}([a, b]) = b - a, \quad \text{and} \quad \mathbb{P}(A) \text{ is unchanged when } A \text{ is slid along the interval.}$$

No function on all subsets of $[0, 1]$ has both.¹ Since we are unwilling to give up either property, the domain of $\mathbb{P}$ must be smaller than 2^S : a class of subsets still large enough

¹Vitali's construction. Call $x, y \in [0, 1]$ equivalent when $x - y \in \mathbb{Q}$, and pick one representative from each equivalence class, collecting them into a set V . Adding each rational $q \in [0, 1)$ to V , modulo 1, produces countably many *disjoint* sets whose union is all of $[0, 1]$. Shift-invariance forces every one of them to have the same probability p , and countable additivity then makes the total 0 if $p = 0$ and ∞ if $p > 0$, never 1. So no probability can be assigned to V at all.

to contain every event we care about, and closed under the operations we perform on events.

Definition 1.3 (σ -algebra). A collection $\mathcal{B}$ of subsets of S is a **σ -algebra** if

- (a) $\emptyset \in \mathcal{B}$;
- (b) $A \in \mathcal{B} \implies A^c \in \mathcal{B}$;
- (c) $A_1, A_2, \dots \in \mathcal{B} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$.

From (a)–(b), $S = \emptyset^c \in \mathcal{B}$; closure under countable intersections is then free, since $\bigcap_i A_i = (\bigcup_i A_i^c)^c$. The word *countable* in (c) is essential: closure under uncountable unions is exactly what Definition 1.3 must avoid. Standard choices: $\mathcal{B} = \{\emptyset, S\}$ (trivial); $\mathcal{B} = 2^S$, all subsets, which is the default for countable S and has 2^n elements when $\#(S) = n$; and, for $S = \mathbb{R}$, the collection generated by the intervals (the *Borel sets*, Definition 1.5 below).

Proposition 1.4 (Intersections of σ -algebras). *If $\{\mathcal{B}_\alpha : \alpha \in \Gamma\}$ is a nonempty family of σ -algebras on S , then $\bigcap_\alpha \mathcal{B}_\alpha$ is a σ -algebra on S .*

Proof. Each of the three properties holds in every $\mathcal{B}_\alpha$ separately, hence in the intersection: $\emptyset$ lies in all; if A lies in all so does A^c ; if $A_1, A_2, \dots$ lie in all so does $\bigcup_i A_i$. $\square$

Definition 1.5 (Generated σ -algebra; Borel sets). For a collection $\mathcal{C}$ of subsets of S , $\sigma(\mathcal{C})$, the σ -algebra **generated by $\mathcal{C}$** , is the intersection of all σ -algebras containing $\mathcal{C}$. The **Borel σ -algebra** $\mathcal{B}(\mathbb{R})$ is the one generated by the open sets of $\mathbb{R}$.

This is well posed: $2^S \supset \mathcal{C}$ is one such σ -algebra, so the family intersected is nonempty, and Proposition 1.4 makes the intersection a σ -algebra. By construction $\sigma(\mathcal{C})$ is the smallest σ -algebra containing $\mathcal{C}$.

Remark 1.6 (Many generators, one σ -algebra). $\mathcal{B}(\mathbb{R})$ is generated by each of

$$\{(a, b)\}, \quad \{(a, b]\}, \quad \{[a, b]\}, \quad \{(-\infty, a] : a \in \mathbb{R}\}, \quad \{(a, b) : a, b \in \mathbb{Q}\}.$$

The last is the informative one: every open $U \subset \mathbb{R}$ is a *countable* union of rational-endpoint intervals. So $\mathcal{B}(\mathbb{R})$ is generated by a countable class, which is why it is small enough to carry a probability measure while $2^{\mathbb{R}}$ is not. We use the generator $\{(-\infty, a]\}$ in Section 2.1 to check measurability cheaply.

1.2 Probability measures

Definition 1.7 (Probability measure; probability space). Given S and a σ -algebra $\mathcal{B}$, a **probability measure** is a function $\mathbb{P}$ on $\mathcal{B}$ with

(A1) $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{B}$;

(B2) $\mathbb{P}(S) = 1$;

(C3) $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$ for pairwise disjoint $A_1, A_2, \dots \in \mathcal{B}$. (countable additivity)

The triple $(S, \mathcal{B}, \mathbb{P})$ is a **probability space**.

These are the **Kolmogorov axioms**. They do not say which $\mathbb{P}$ to choose: for a coin they give only $\mathbb{P}(\{H\}) + \mathbb{P}(\{T\}) = 1$. Calling the coin *fair* adds the symmetry judgement $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\})$ from outside the axioms, and only then does $\frac{1}{2}$ follow; assigning $\frac{1}{9}$ and $\frac{8}{9}$ is equally legitimate and describes a different coin.

Theorem 1.8 (Probabilities on a countable space). *Let $S = \{s_1, s_2, \dots\}$ be finite or countable, $\mathcal{B}$ any σ -algebra on S , and $p_1, p_2, \dots \geq 0$ with $\sum_i p_i = 1$. Then*

$$\mathbb{P}(A) = \sum_{\{i: s_i \in A\}} p_i \quad (3)$$

(empty sum = 0) defines a probability measure on $\mathcal{B}$.

Proof. We check the three axioms in turn.

(A1) $\mathbb{P}(A)$ is a sum of some subcollection of the p_i , each of which is nonnegative by hypothesis, so $\mathbb{P}(A) \geq 0$.

(B2) Every s_i lies in S , so the index set in (3) is the whole of $\{1, 2, \dots\}$ and $\mathbb{P}(S) = \sum_i p_i = 1$ by hypothesis.

(C3) Let $A_1, A_2, \dots \in \mathcal{B}$ be pairwise disjoint and put $A = \bigcup_{k=1}^{\infty} A_k$. By (3),

$$\mathbb{P}(A) = \sum_{\{i: s_i \in A\}} p_i.$$

Now $s_i \in A$ if and only if $s_i \in A_k$ for at least one k , and disjointness makes that

k unique. Hence the index set $\{i : s_i \in A\}$ is the *disjoint* union over k of the index sets $\{i : s_i \in A_k\}$, so splitting the sum along that decomposition gives

$$\sum_{\{i : s_i \in A\}} p_i = \sum_{k=1}^{\infty} \sum_{\{i : s_i \in A_k\}} p_i = \sum_{k=1}^{\infty} \mathbb{P}(A_k).$$

□

So on a countable space, specifying $\mathbb{P}$ amounts to specifying the point masses $p_i = \mathbb{P}(\{s_i\})$. On an uncountable space that route is closed, because individual points typically carry no probability at all. Take $S = [0, 1]$ with the uniform probability, for which an interval receives its length. Fix $s \in [0, 1]$. For every $\varepsilon > 0$ we have $\{s\} \subset [s - \varepsilon, s + \varepsilon] \cap [0, 1]$, so monotonicity (Theorem 1.11(c)) gives

$$0 \leq \mathbb{P}(\{s\}) \leq \mathbb{P}([s - \varepsilon, s + \varepsilon] \cap [0, 1]) \leq 2\varepsilon.$$

Since this holds for *every* $\varepsilon > 0$, we get $\mathbb{P}(\{s\}) = 0$. Under any $\mathbb{P}$, at most n points can satisfy $\mathbb{P}(\{s\}) > 1/n$: if there were $n + 1$ of them, those singletons would be disjoint and (A3) would give them total probability greater than $(n + 1)/n > 1$. The set of points of positive probability is therefore $\bigcup_{n=1}^{\infty} \{s : \mathbb{P}(\{s\}) > 1/n\}$, a countable union of finite sets, hence countable. When S is uncountable, all but countably many of its points are invisible to $\mathbb{P}$, so listing point masses cannot describe the measure.

Example 1.9 (Throwing a die until a six). Throw a die repeatedly until a 6 appears. An outcome is a whole sequence, so $S = \{1, \dots, 6\}^{\mathbb{N}}$ is uncountable and 2^S unusable. Take $\mathcal{B} = \sigma(\{C_{i_1 \dots i_n}\})$ generated by the **cylinder sets**

$$C_{i_1 \dots i_n} = \{s \in S : s_1 = i_1, \dots, s_n = i_n\},$$

which formalize “the first n throws came out thus”. On these we declare

$$\mathbb{P}(C_{i_1 \dots i_n}) = \frac{1}{6^n} \quad \text{for every } n \text{ and every } (i_1, \dots, i_n), \quad (4)$$

which says the die is fair and the throws independent.

This prescription covers only the cylinders, not all of $\mathcal{B}$. That it extends, and extends in only one way, is the content of the **Kolmogorov extension theorem**: an assignment of probabilities to the cylinder sets extends to a unique probability

measure on the generated σ -algebra provided it is **consistent**, meaning that the numbers attached to cylinders of length $n + 1$ sum, over the free last coordinate, to the number attached to the corresponding cylinder of length n . Here consistency is the identity $6 \times 6^{-(n+1)} = 6^{-n}$. The lesson generalizes: it is neither necessary nor usually possible to specify $\mathbb{P}$ on all of $\mathcal{B}$, and a consistent specification on a generating class suffices.

Now compute the probability that the first 6 appears on throw k . That event is

$$E_k = \{s \in S : s_1 \neq 6, \dots, s_{k-1} \neq 6, s_k = 6\},$$

and it is a *finite* disjoint union of cylinders of length k : one for each way of filling the first $k - 1$ coordinates with values in $\{1, \dots, 5\}$, with the k th coordinate fixed at 6. There are 5^{k-1} such cylinders, they are disjoint because they differ in some coordinate, and each has probability 6^{-k} by (4). So (A3) gives

$$\mathbb{P}(E_k) = 5^{k-1} \cdot \frac{1}{6^k} = \left(\frac{5}{6}\right)^{k-1} \left(\frac{1}{6}\right), \quad k = 1, 2, \dots$$

Reading the two factors: $k - 1$ throws must fail, each with probability $\frac{5}{6}$, and the k th must succeed, with probability $\frac{1}{6}$. A random variable X taking values in $\{1, 2, \dots\}$ with

$$\mathbb{P}(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, \dots, \quad (5)$$

is said to have the **geometric distribution** with parameter p ; it counts the number of independent trials up to and including the first success, where each trial succeeds with probability p . Here $p = \frac{1}{6}$. As a check that no probability has been lost, the geometric series sums to $\sum_{k \geq 1} (1 - p)^{k-1} p = p/[1 - (1 - p)] = 1$.

Theorem 1.10 (Elementary properties). *For any $A \in \mathcal{B}$,*

- (a) $\mathbb{P}(\emptyset) = 0$;
- (b) $\mathbb{P}(A) \leq 1$;
- (c) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Proof. A, A^c are disjoint with union S , so $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$, which is (c); (b) follows since $\mathbb{P}(A^c) \geq 0$. For (a), $S = S \cup \emptyset$ is disjoint, so $1 = \mathbb{P}(S) + \mathbb{P}(\emptyset)$. $\square$

Theorem 1.11 (Two-event calculus). *For $A, B \in \mathcal{B}$:*

- (a) $\mathbb{P}(B \cap A^c) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
- (b) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
- (c) $A \subset B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$. (*monotonicity*)

Proof. Item by item:

- (a) By (1), $\mathbb{P}(B) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap A^c)$.
- (b) By (2), $A \cup B = A \cup (B \cap A^c)$ is disjoint, so with (a),

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \quad (6)$$

- (c) If $A \subset B$ then $A \cap B = A$, so (a) gives $0 \leq \mathbb{P}(B) - \mathbb{P}(A)$.

□

Since $\mathbb{P}(A \cup B) \leq 1$, (6) rearranges to **Bonferroni's inequality**,

$$\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1, \quad (7)$$

which bounds an intersection using only the individual probabilities. It is useful only when those are large: at $\mathbb{P}(A) = \mathbb{P}(B) = .95$ it gives $\mathbb{P}(A \cap B) \geq .90$, but at .4 it gives the correct and vacuous ≥ -0.2 .

Theorem 1.12. *Collections of events*

- (a) $\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A \cap C_i)$ for any partition $C_1, C_2, \dots$ of S ; (*law of total probability*)
- (b) $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$. (*Boole's inequality*)

Proof. Item by item:

- (a) By distributivity $A = \bigcup_i (A \cap C_i)$, and these are pairwise disjoint because the C_i are; apply (A3).
- (b) Disjointify: $A_1^* = A_1$ and $A_i^* = A_i \setminus \bigcup_{j < i} A_j$ for $i \geq 2$. Then $\bigcup_i A_i^* = \bigcup_i A_i$, and

the A_i^* are pairwise disjoint: for $i > k$, A_i^* contains the factor A_k^c while $A_k^* \subset A_k$. Since $A_i^* \subset A_i$, monotonicity gives $\mathbb{P}(A_i^*) \leq \mathbb{P}(A_i)$, so

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i^*) \leq \sum_i \mathbb{P}(A_i).$$

□

Corollary 1.13 (Bonferroni, general form). $\mathbb{P}(\bigcap_{i=1}^n A_i) \geq \sum_{i=1}^n \mathbb{P}(A_i) - (n-1)$.

Proof. Boole applied to the complements gives $\mathbb{P}(\bigcup_i A_i^c) \leq \sum_i \mathbb{P}(A_i^c)$; using $\bigcup_i A_i^c = (\bigcap_i A_i)^c$ and Theorem 1.10(c) this reads $1 - \mathbb{P}(\bigcap_i A_i) \leq n - \sum_i \mathbb{P}(A_i)$. □

So Boole and Bonferroni are one statement seen from two sides, and both are truncations of the **inclusion–exclusion identity**: with $P_j = \sum_{i_1 < \dots < i_j} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_j})$,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = P_1 - P_2 + P_3 - \dots \pm P_n, \quad (8)$$

whose partial sums alternately over- and under-state the left side. For $n = 2$ this is Theorem 1.11(b), and truncating after P_1 is Boole.

Proposition 1.14 (Continuity of probability). *If $A_1 \subset A_2 \subset \dots$ with $A = \bigcup_n A_n$, or $A_1 \supset A_2 \supset \dots$ with $A = \bigcap_n A_n$, then $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$. In particular $A_n \downarrow \emptyset$ implies $\mathbb{P}(A_n) \rightarrow 0$.*

Proof. Increasing case. The sets A_n are not disjoint, so (A3) does not apply to them directly. Replace them by the increments

$$B_1 = A_1, \quad B_i = A_i \setminus A_{i-1} \quad (i \geq 2),$$

each of which lies in $\mathcal{B}$. Three facts:

- (i) *The B_i are pairwise disjoint.* Let $i < k$. Then $B_k \subset A_{k-1}^c \subset A_i^c$, the second containment because $A_i \subset A_{k-1}$ by increasingness; and $B_i \subset A_i$. So $B_i \cap B_k = \emptyset$.
- (ii) $\bigcup_{i=1}^n B_i = A_n$. Each $B_i \subset A_i \subset A_n$ for $i \leq n$, giving “ $\subset$ ”. Conversely, if $x \in A_n$, let i be the smallest index with $x \in A_i$; then $x \notin A_{i-1}$ (or $i = 1$), so $x \in B_i$ with $i \leq n$.

(iii) $\bigcup_{i=1}^{\infty} B_i = A$. Take the union over all n in (ii) and use $A = \bigcup_n A_n$.

Now apply (A3) to the B_i , then use the definition of an infinite series and (ii):

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(B_i) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{i=1}^n B_i\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n),$$

the fourth equality by finite additivity applied to the disjoint $B_1, \dots, B_n$.

Decreasing case. If $A_n \downarrow A$ then $A_n^c \uparrow A^c$: the complements increase because the A_n decrease, and $\bigcup_n A_n^c = (\bigcap_n A_n)^c = A^c$ by DeMorgan. The increasing case gives $\mathbb{P}(A_n^c) \rightarrow \mathbb{P}(A^c)$, and Theorem 1.10(c) turns this into $1 - \mathbb{P}(A_n) \rightarrow 1 - \mathbb{P}(A)$, that is, $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$.

Finally, $A_n \downarrow \emptyset$ is the special case $A = \emptyset$, where $\mathbb{P}(A) = 0$ by Theorem 1.10(a). $\square$

1.3 Counting

When S is finite with all outcomes equally likely, Theorem 1.8 fixes the $p_i = 1/N$, and by (A3)

$$\mathbb{P}(A) = \frac{\text{number of elements in } A}{\text{number of elements in } S}. \quad (9)$$

Computing a probability then reduces to counting, governed by the following.

Theorem 1.15 (Fundamental theorem of counting). *If a job consists of k tasks, the i th doable in n_i ways, the job can be done in $n_1 \times \dots \times n_k$ ways.*

Proof. For $k = 2$, grouping outcomes by the choice at the first task partitions them into n_1 blocks of size n_2 . The general case follows by induction. $\square$

Recall $n! = n(n-1) \dots 1$ with $0! = 1$, and $\binom{n}{r} = n!/[r!(n-r)!]$. Any “draw r from n ” problem is then settled by two questions: is a drawn object replaced, and does order matter? The four answers give Table 1. Three of the four cells are immediate from Theorem 1.15: ordered without replacement gives $n(n-1) \dots (n-r+1)$; ordered with replacement gives n^r ; and unordered without replacement divides the first by the $r!$ redundant orderings.

Table 1: Number of arrangements of size r from n objects.

	Without replacement	With replacement
Ordered	$\frac{n!}{(n-r)!}$	n^r
Unordered	$\binom{n}{r}$	$\binom{n+r-1}{r}$

The fourth cell is the one worth deriving. Unordered with replacement is *not* $n^r/r!$, which need not even be an integer. Change representation. In the “stars and bars” picture we treat the n values as bins and place r markers,

$$\bullet \mid \mid \bullet \bullet \mid \mid \cdots \mid \bullet \mid \bullet \bullet \mid .$$

Only the arrangement of markers and *interior* walls matters; n bins give $n+1$ walls, of which the two outermost are uninformative, leaving $n-1$. So we arrange $n-1+r$ objects, r indistinguishable markers and $n-1$ indistinguishable walls, in $\binom{n+r-1}{r}$ ways. The bookkeeping for repeated values is the **multinomial coefficient**: k places filled by m distinct values repeated $k_1, \dots, k_m$ times admit

$$\frac{k!}{k_1! k_2! \cdots k_m!}, \quad k_1 + \cdots + k_m = k, \quad (10)$$

orderings: the $k!$ arrangements of distinguishable objects, divided by the permutations within each repeated group. Drawing four numbers with replacement from $\{2, 4, 9, 12\}$, for instance, the sample $\{2, 4, 4, 9\}$ has $4!/2! = 12$ ordered components and so probability $12/4^4 = 12/256$, not the $1/35$ that equal weighting of the $\binom{7}{4} = 35$ unordered samples would give. Computing the induced distribution of the sample average this way is an elementary instance of the **bootstrap** [Efron and Tibshirani, 1993].

1.4 Conditional probability and independence

Definition 1.16 (Conditional probability). If $\mathbb{P}(B) > 0$, the **conditional probability of A given B** is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (11)$$

B becomes the sample space: $\mathbb{P}(B \mid B) = 1$, and all further events are calibrated against their overlap with B . Moreover $\mathbb{P}(\cdot \mid B)$ is itself a probability measure. (A1) and (A2) are immediate from (11), and (A3) follows from $(\bigcup_i A_i) \cap B = \bigcup_i (A_i \cap B)$. So the entire calculus of Section 1.2 applies to it verbatim, a fact we use without comment. Rearranging (11) gives the **multiplication rule** and, iterated, the **chain rule**:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A), \quad (12)$$

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}). \quad (13)$$

Equating the two right-hand sides of (12) turns a conditional around, $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A)\mathbb{P}(A)/\mathbb{P}(B)$.

Theorem 1.17 (Bayes' rule). *Let $A_1, A_2, \dots$ partition S and let $\mathbb{P}(B) > 0$. Then*

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_j \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}. \quad (14)$$

Proof. The numerator is $\mathbb{P}(A_i \cap B)$ by (12); the denominator is $\mathbb{P}(B) = \sum_j \mathbb{P}(B \cap A_j)$ by Theorem 1.12(a). $\square$

Read (14) with A_i a parameter value and B the data. The denominator does not depend on A_i , so the content of the theorem is that the posterior is proportional to the likelihood times the prior.

Example 1.18 (Three prisoners). Prisoners A, B, C are on death row; the governor pardons one at random and tells the warden. A asks which of B, C will die and the warden says B . A reasons that his chance has risen to $\frac{1}{2}$. He is wrong.

Let A, B, C also denote the pardon events, each of probability $\frac{1}{3}$, and let W be the event that the warden says B will die. If A is pardoned the warden names B or C with equal probability; otherwise he must name whoever is not pardoned. So

$$\mathbb{P}(W) = \underbrace{\frac{1}{2} \cdot \frac{1}{3}}_{A \text{ pardoned}} + \underbrace{0}_B + \underbrace{1 \cdot \frac{1}{3}}_C = \frac{1}{2}, \quad \mathbb{P}(A \mid W) = \frac{1/6}{1/2} = \frac{1}{3}.$$

This is the Monty Hall problem in disguise; see Morgan et al. [1991].

Sometimes learning B changes nothing, $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. By (12) this is equivalent to $\mathbb{P}(B \mid A) = \mathbb{P}(B)$, and both are equivalent to a symmetric statement free of positivity requirements.

Definition 1.19 (Independence). Events A, B are **statistically independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B). \quad (15)$$

Events $A_1, \dots, A_n$ are **mutually independent** if for *every* subcollection $A_{i_1}, \dots, A_{i_k}$,

$$\mathbb{P}\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k \mathbb{P}(A_{i_j}). \quad (16)$$

Independence propagates to complements: if A, B are independent then so are A, B^c , since by Theorem 1.11(a)

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) [1 - \mathbb{P}(B)] = \mathbb{P}(A)\mathbb{P}(B^c),$$

and likewise for the remaining pairs. Mutual independence, by contrast, is $2^n - n - 1$ separate equations, and no proper subset of them suffices.

2 Lecture 2: Random Variables and Distributions

2.1 Random variables

It is usually easier to work with a summary of the outcome than with the outcome itself. Poll 50 people recording 1 for agree and 0 for disagree: the sample space has $2^{50} \approx 1.1 \times 10^{15}$ points. If all we care about is how many agree, setting $X =$ number of 1s replaces it by $\{0, 1, \dots, 50\}$. That reduction is what a random variable is for.

Definition 2.1 (Random variable). Let $(S, \mathcal{B}, \mathbb{P})$ be a probability space. A function $X : S \rightarrow \mathbb{R}$ is a **random variable** if it is **$\mathcal{B}$ -measurable**:

$$X^{-1}(A) = \{s \in S : X(s) \in A\} \in \mathcal{B} \quad \text{for every } A \in \mathcal{B}(\mathbb{R}). \quad (17)$$

Proposition 2.2 (Measurability on a generating class). *Let $\mathcal{C}$ satisfy $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$. Then X is measurable if and only if $X^{-1}(C) \in \mathcal{B}$ for every $C \in \mathcal{C}$. In particular X is a random variable if and only if*

$$\{s \in S : X(s) \leq a\} \in \mathcal{B} \quad \text{for every } a \in \mathbb{R}. \quad (18)$$

Proof. Necessity is immediate. For sufficiency let $\mathcal{A} = \{A \subset \mathbb{R} : X^{-1}(A) \in \mathcal{B}\}$. Preimages commute with complements and unions, so $\mathcal{A}$ inherits the three properties of Definition 1.3 from $\mathcal{B}$ and is a σ -algebra. Since $\mathcal{C} \subset \mathcal{A}$, minimality of $\sigma(\mathcal{C})$ gives $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subset \mathcal{A}$. The second claim is the case $\mathcal{C} = \{(-\infty, a]\}$, a generator by Remark 1.6. $\square$

The basic example is the **indicator** $\mathbf{1}\{A\}$ of an event $A \in \mathcal{B}$, equal to 1 on A and 0 off it: $\mathbf{1}\{A\}^{-1}(C)$ is one of $\emptyset, A, A^c, S$, all in $\mathcal{B}$. Indicators bridge events and random variables through $\mathbb{P}(A) = \mathbb{E}[\mathbf{1}\{A\}]$, once expectations are available.

2.2 The induced probability

Defining X also defines a new sample space, its range $\mathcal{X}$. We must check that $\mathbb{P}$ transfers. We observe $X \in A$ exactly when the outcome lies in $X^{-1}(A)$, so define

$$\mathbb{P}_X(X \in A) = \mathbb{P}(\{s \in S : X(s) \in A\}) = \mathbb{P}(X^{-1}(A)), \quad A \in \mathcal{B}(\mathbb{R}). \quad (19)$$

Theorem 2.3 (Induced probability space). *If X is a random variable on $(S, \mathcal{B}, \mathbb{P})$, then $\mathbb{P}_X = \mathbb{P} \circ X^{-1}$ is a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, so $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_X)$ is a probability space.*

Proof. Measurability makes $\mathbb{P}_X$ well defined on $\mathcal{B}(\mathbb{R})$.

(A1) $\mathbb{P}_X(A) = \mathbb{P}(X^{-1}(A)) \geq 0$, since $\mathbb{P}$ satisfies (A1).

(B2) $X^{-1}(\mathbb{R}) = S$, so $\mathbb{P}_X(\mathbb{R}) = \mathbb{P}(S) = 1$.

(C3) Preimages of disjoint sets are disjoint, since $s \in X^{-1}(A_i) \cap X^{-1}(A_k)$ would give $X(s) \in A_i \cap A_k = \emptyset$, and preimages commute with unions, so

$$\mathbb{P}_X\left(\bigcup_i A_i\right) = \mathbb{P}\left(\bigcup_i X^{-1}(A_i)\right) = \sum_i \mathbb{P}(X^{-1}(A_i)) = \sum_i \mathbb{P}_X(A_i),$$

the middle step by (A3) for $\mathbb{P}$.

□

$\mathbb{P}_X$ is the **induced probability**, the **distribution of X** , or the **pushforward** of $\mathbb{P}$ under X . By the equivalence in (19) we write $\mathbb{P}(X \in A)$ rather than $\mathbb{P}_X(A)$. Random variables are denoted by uppercase letters and their realized values by the corresponding lowercase letter.

This is the first genuinely consequential move of the course. An experiment $(S, \mathcal{B}, \mathbb{P})$ of arbitrary complexity has been replaced by a probability space on the *real line*, and from here S never reappears: estimators, test statistics, and regression coefficients are all random variables in this sense.

Example 2.4 (Three coin tosses). Let X be the number of heads in three tosses of a fair coin:

s	HHH	HHT	HTH	THH	HTT	THT	TTH	TTT
$X(s)$	3	2	2	2	1	1	1	0

Each sample point has probability $\frac{1}{8}$, so $\mathbb{P}_X$ assigns $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ to $x = 0, 1, 2, 3$; for instance $\mathbb{P}(X = 1) = \mathbb{P}(\{HTT, THT, TTH\}) = \frac{3}{8}$. Note that X collapses eight outcomes to four values: it is not injective, and the loss of information is deliberate.

Remark 2.5 (The uncountable case). When $\mathcal{X}$ is countable, listing $\mathbb{P}(X = x)$ determines $\mathbb{P}_X$. When $\mathcal{X}$ is uncountable this fails, since typically $\mathbb{P}(X = x) = 0$ for every

x , and (19) becomes the operative definition. This is the situation of Theorem 1.8 transplanted from S to $\mathcal{X}$.

2.3 Distribution functions

How do we describe $\mathbb{P}_X$ when $\mathcal{X}$ is uncountable? Proposition 2.2 points the way: the sets $(-\infty, x]$ generate $\mathcal{B}(\mathbb{R})$, so knowing $\mathbb{P}_X$ on them determines it everywhere.

Definition 2.6 (Cumulative distribution function). The **cumulative distribution function**, or **cdf**, of a random variable X , denoted F_X , is

$$F_X(x) = \mathbb{P}_X(X \leq x) = \mathbb{P}(\{s \in S : X(s) \leq x\}) = \mathbb{P}_X((-\infty, x]), \quad x \in \mathbb{R}. \quad (20)$$

Example 2.7 (Tossing three coins). Take X = number of heads in three tosses of a fair coin, as in Example 2.4, where $\mathbb{P}_X$ put mass $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ on 0, 1, 2, 3. Accumulating,

$$F_X(x) = \begin{cases} 0 & -\infty < x < 0, \\ \frac{1}{8} & 0 \leq x < 1, \\ \frac{1}{2} & 1 \leq x < 2, \\ \frac{7}{8} & 2 \leq x < 3, \\ 1 & 3 \leq x < \infty, \end{cases} \quad (21)$$

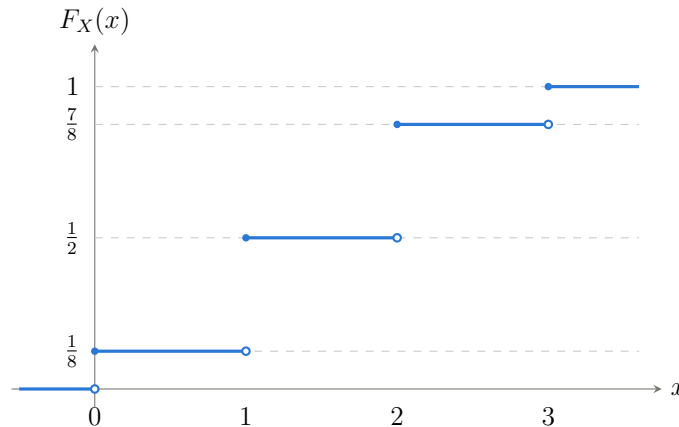


Figure 1: Cdf of Example 2.7. The function is a step function, constant between the integers and jumping at each of them.

Figure 1 shows that a cdf may be discontinuous. Where it jumps, the definition $F_X(x) = \mathbb{P}(X \leq x)$ with a weak inequality makes F_X take the value at the top of the jump: the function is continuous when the point is approached from the right, a property called **right-continuity**. Had we instead defined $F_X(x) = \mathbb{P}(X < x)$ with a strict inequality, F_X would have been left-continuous. Right-continuity is therefore not an assumption but a consequence of Definition 2.6.

Theorem 2.8 (Properties of a cdf). *A function $F : \mathbb{R} \rightarrow [0, 1]$ is the cdf of some random variable if and only if*

- (a) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$;
- (b) F is nondecreasing;
- (c) F is right-continuous, that is, $\lim_{x \downarrow x_0} F(x) = F(x_0)$ for every x_0 .

Proof. We prove necessity, which is where the probability lies; sufficiency is discussed after the proof.

(a) Let $x_n \downarrow -\infty$. The events $A_n = \{X \leq x_n\}$ decrease with $\bigcap_n A_n = \emptyset$, since no real number is below every x_n . Continuity from above (Proposition 1.14) gives $F(x_n) = \mathbb{P}(A_n) \rightarrow \mathbb{P}(\emptyset) = 0$. Similarly, if $x_n \uparrow \infty$ then $A_n = \{X \leq x_n\}$ increases with $\bigcup_n A_n = \mathbb{R}$, so continuity from below gives $F(x_n) \rightarrow 1$. As the limits agree along every such sequence, the limits in x exist and equal 0 and 1.

(b) If $x \leq y$ then $\{X \leq x\} \subset \{X \leq y\}$, so monotonicity of $\mathbb{P}$ (Theorem 1.11(c)) gives $F(x) \leq F(y)$.

(c) Let $x_n \downarrow x_0$. The events $\{X \leq x_n\}$ decrease and

$$\bigcap_{n=1}^{\infty} \{X \leq x_n\} = \{X \leq x_0\},$$

because $X(s) \leq x_n$ for every n forces $X(s) \leq \lim_n x_n = x_0$, and conversely. Continuity from above gives $F(x_n) \rightarrow F(x_0)$. $\square$

Sufficiency, that any F with (a)–(c) is the cdf of some random variable, is harder: one must exhibit a probability space $(S, \mathcal{B}, \mathbb{P})$ and a random variable X on it with that cdf. The standard construction takes $S = (0, 1)$ with Lebesgue measure, itself built from outer measure, and $X(s) = \inf\{x : F(x) \geq s\}$, the generalized inverse of F . Note that $(0, 1)$ carrying Lebesgue measure is exactly the uniform distribution $U[0, 1]$. We omit the verification.

Example 2.9 (Tossing until a head). Toss a coin until a head appears, with p the probability of a head on any toss, and let X be the number of tosses required. As in Example 1.9, getting the first head on toss x means $x - 1$ tails followed by a head, so

$$\mathbb{P}(X = x) = (1 - p)^{x-1}p, \quad x = 1, 2, \dots \quad (22)$$

Summing and using the partial sum of the geometric series, $\sum_{k=1}^n t^{k-1} = (1 - t^n)/(1 - t)$ for $t \neq 1$,

$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{i=1}^x (1 - p)^{i-1}p = \frac{1 - (1 - p)^x}{1 - (1 - p)} p = 1 - (1 - p)^x, \quad x = 1, 2, \dots,$$

and F_X is flat between consecutive integers, as in Example 2.7. It is graphed in Figure 2 for $p = .3$.

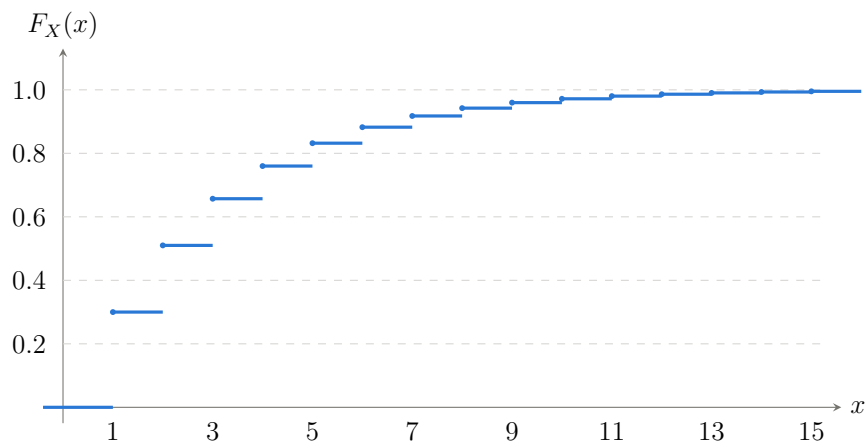


Figure 2: Geometric cdf of Example 2.9, $p = .3$: $F_X(x) = 1 - (1 - p)^{\lfloor x \rfloor}$. The jump at the integer k has height $(1 - p)^{k-1}p$, so the jumps shrink geometrically and F_X approaches 1 without ever reaching it.

Example 2.10 (A continuous cdf). The function

$$F_X(x) = \frac{1}{1 + e^{-x}}, \quad x \in \mathbb{R}, \quad (23)$$

satisfies Theorem 2.8: $e^{-x} \rightarrow \infty$ as $x \rightarrow -\infty$ gives $F_X \rightarrow 0$, and $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$ gives $F_X \rightarrow 1$; differentiating,

$$\frac{d}{dx}F_X(x) = \frac{e^{-x}}{(1 + e^{-x})^2} > 0,$$

so F_X is increasing; and F_X is not merely right-continuous but continuous. This is the **logistic** distribution.

Example 2.11 (A cdf with a jump). A cdf need not be either a step function or continuous: it can mix the two. Modify (23) by inserting a jump of height $\varepsilon \in (0, 1)$ at the origin,

$$F_Y(y) = \begin{cases} \frac{1 - \varepsilon}{1 + e^{-y}} & y < 0, \\ \varepsilon + \frac{1 - \varepsilon}{1 + e^{-y}} & y \geq 0, \end{cases} \quad (24)$$

graphed in Figure 3. This F_Y is continuous everywhere except at 0, where it jumps by ε , and it is right-continuous there because the second branch is the one that applies at $y = 0$.

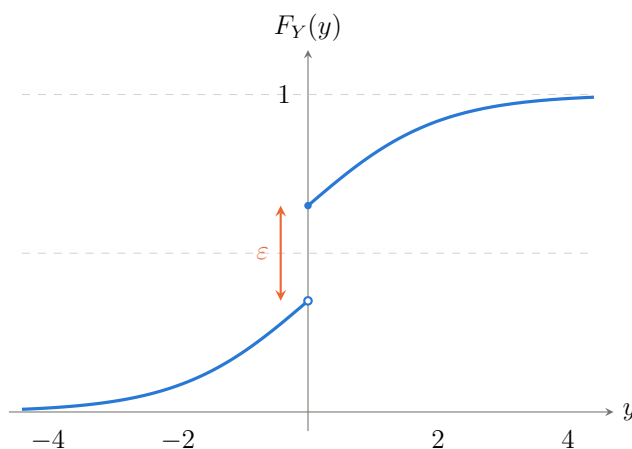


Figure 3: The mixed cdf (24) of Example 2.11, drawn with $\varepsilon = .3$. It is continuous away from the origin and jumps there by ε , so Y is neither discrete nor continuous.

Definition 2.12 (Continuous and discrete random variables). A random variable X is **discrete** if its cdf accumulates a mass function,

$$F_X(x) = \sum_{t \leq x} f_X(t) \quad (25)$$

for a probability mass function f_X ; and **continuous** if its cdf integrates a density,

$$F_X(x) = \int_{-\infty}^x f_X(w) dw \quad (26)$$

for some function f_X .

Condition (26) is strictly stronger than continuity of F_X : it says that F_X is **absolutely continuous**, that is, F'_X exists almost everywhere, is Lebesgue integrable, and recovers F_X by integration,

$$F_X(t) = F_X(a) + \int_a^t F'_X(s) ds \quad \text{for all } a < t. \quad (27)$$

A continuous but not absolutely continuous cdf exists, the **Cantor function** is the standard example.

Definition 2.13 (Identically distributed). Random variables X and Y are **identically distributed** if $\mathbb{P}(X \in A) = \mathbb{P}(Y \in A)$ for every $A \in \mathcal{B}(\mathbb{R})$.

Example 2.14 (Identically distributed but not equal). Toss a fair coin three times as in Example 2.4 and set

$$X = \text{number of heads observed}, \quad Y = \text{number of tails observed}.$$

By the symmetry of the fair coin, $\mathbb{P}(X = k) = \mathbb{P}(Y = k) = \binom{3}{k} \frac{1}{8}$ for $k = 0, 1, 2, 3$, so X and Y are identically distributed. Yet $X(s) = Y(s)$ for *no* sample point s , since $X(s) + Y(s) = 3$ would force $X(s) = \frac{3}{2}$, which is not an integer. Identically distributed does not mean equal.

Theorem 2.15 (Correspondence theorem). *The following are equivalent.*

- (a) X and Y are identically distributed.
- (b) $F_X(x) = F_Y(x)$ for every x .

Proof. (a) $\implies$ (b): for each x the set $(-\infty, x]$ is Borel, so

$$F_X(x) = \mathbb{P}(X \in (-\infty, x]) = \mathbb{P}(Y \in (-\infty, x]) = F_Y(x).$$

(b) $\implies$ (a) is the substantive direction: knowing that the two distributions agree on the intervals $(-\infty, x]$, we must conclude that they agree on *every* Borel set. This is where Remark 1.6 pays off. The sets $(-\infty, x]$ generate $\mathcal{B}(\mathbb{R})$, and a standard uniqueness theorem for measures says that two probability measures agreeing on a generating class which is closed under finite intersections agree on the whole generated σ -algebra. The class $\{(-\infty, x] : x \in \mathbb{R}\}$ is closed under intersections, since $(-\infty, x] \cap (-\infty, y] = (-\infty, x \wedge y]$, so the theorem applies [Billingsley, 1995, §3]. $\square$

2.4 Density and mass functions

Definition 2.16 (Probability mass function). The **probability mass function** (**pmf**) of a discrete random variable X is $f_X(x) = \mathbb{P}(X = x)$ for all x .

For the geometric variable of Example 2.9,

$$f_X(x) = \begin{cases} (1-p)^{x-1}p & x = 1, 2, \dots \\ 0 & \text{otherwise,} \end{cases}$$

and $f_X(x)$ is the size of the jump of F_X at x . Because single points now carry probability, event probabilities are sums: for integers $a \leq b$,

$$\mathbb{P}(a \leq X \leq b) = \sum_{k=a}^b f_X(k), \quad \text{and in particular} \quad F_X(b) = \sum_{k=1}^b f_X(k).$$

Proposition 2.17. *If X is a continuous random variable then $\mathbb{P}(X = x) = 0$ for every x .*

Proof. For any $\varepsilon > 0$ we have $\{X = x\} \subset \{x - \varepsilon < X \leq x\}$, so monotonicity (Theorem 1.11(c)) gives

$$0 \leq \mathbb{P}(X = x) \leq \mathbb{P}(x - \varepsilon < X \leq x) = F_X(x) - F_X(x - \varepsilon).$$

Letting $\varepsilon \downarrow 0$, the right side tends to 0 by continuity of F_X . $\square$

Definition 2.18 (Probability density function). The **probability density function** (**pdf**) of a continuous random variable X is a function f_X satisfying

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \quad \text{for all } x. \quad (28)$$

Since F_X is nondecreasing it is differentiable almost everywhere, and there

$$\frac{d}{dx} F_X(x) = f_X(x) \quad \text{a.e.}, \quad (29)$$

so the pdf is recovered from the cdf by differentiation. Changing f_X on a set of measure zero changes no integral, so a density is *essentially unique*: determined only up to a null set. We write $X \sim F_X$, read “ X is distributed as F_X ”, and likewise $X \sim f_X$; if X and Y have the same distribution we write $X \sim Y$. Note that Proposition 2.17 makes the endpoints of an interval irrelevant in the continuous case,

$$\mathbb{P}(a < X < b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X \leq b).$$

Example 2.19 (Logistic density). Differentiating the logistic cdf (23),

$$f_X(x) = \frac{d}{dx} \frac{1}{1 + e^{-x}} = \frac{e^{-x}}{(1 + e^{-x})^2},$$

the bell-shaped curve of Figure 4. Interval probabilities are areas under it:

$$\mathbb{P}(a < X < b) = F_X(b) - F_X(a) = \int_{-\infty}^b f_X(x) dx - \int_{-\infty}^a f_X(x) dx = \int_a^b f_X(x) dx.$$

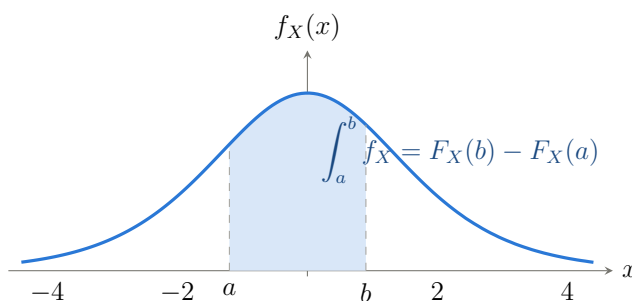


Figure 4: The logistic density of Example 2.19. The shaded area between a and b is $\mathbb{P}(a < X < b)$.

Theorem 2.20 (Characterization of pdfs and pmfs). *A function f_X is the pdf or pmf of some random variable if and only if*

- (a) $f_X(x) \geq 0$ for all x ; that is, $f_X : \mathbb{R} \rightarrow [0, 1]$ in the discrete case and $f_X : \mathbb{R} \rightarrow \mathbb{R}_+$ in the continuous case;
- (b) $\sum_x f_X(x) = 1$ in the discrete case, or $\int_{-\infty}^{\infty} f_X(x) dx = 1$ in the continuous case.

Proof. Necessity. Nonnegativity of a pmf is immediate since $f_X(x) = \mathbb{P}(X = x) \geq 0$; for a pdf it follows from (29), since F_X is nondecreasing by Theorem 2.8(b). For (b) in the continuous case, (28) and Theorem 2.8(a) give

$$1 = \lim_{x \rightarrow \infty} F_X(x) = \int_{-\infty}^{\infty} f_X(t) dt,$$

and the discrete case is the same computation with a sum.

Sufficiency. Given f_X with (a) and (b), define F by (28). Then F is nondecreasing because the integrand is nonnegative, tends to 0 and to 1 at $\mp\infty$ by (b), and is continuous, hence right-continuous. So F satisfies Theorem 2.8 and is the cdf of some random variable, whose pdf is f_X . $\square$

Consequently any $h \geq 0$ vanishing off A with $\int_A h = K \in (0, \infty)$ normalizes into a density $f_X = h/K$; the constant K is the **normalizing constant**.

2.5 Distributions of functions of a random variable

If X is a random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is measurable, then $Y = g(X)$ is again a random variable, and its distribution is determined by that of X together with g . Writing

$$g^{-1}(A) = \{x \in \mathcal{X} : g(x) \in A\}, \quad (30)$$

the argument of Theorem 2.3 applies verbatim and gives

$$\mathbb{P}(Y \in A) = \mathbb{P}(g(X) \in A) = \mathbb{P}(X \in g^{-1}(A)). \quad (31)$$

Throughout we use the **support** of X , $\mathcal{X} = \{x : f_X(x) > 0\}$, and $\mathcal{Y} = \{y : y = g(x) \text{ for some } x \in \mathcal{X}\}$; keeping track of these two sets is where most errors in transformation problems occur.

In the discrete case (31) is all one needs: $\mathcal{Y}$ is countable and

$$f_Y(y) = \mathbb{P}(Y = y) = \sum_{x \in g^{-1}(y)} f_X(x), \quad y \in \mathcal{Y}, \quad (32)$$

so the work is to identify $g^{-1}(y)$ and add up. In the continuous case we go through the cdf,

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \int_{\{x : g(x) \leq y\}} f_X(x) dx, \quad (33)$$

and the difficulty is describing the region $\{x : g(x) \leq y\}$. When g is monotone that region is an interval.

Theorem 2.21 (Monotone transformations, cdf form). *Let $Y = g(X)$ with $\mathcal{X}, \mathcal{Y}$ as above.*

- (a) *If g is increasing on $\mathcal{X}$, then $F_Y(y) = F_X(g^{-1}(y))$ for $y \in \mathcal{Y}$.*
- (b) *If g is decreasing on $\mathcal{X}$ and X is continuous, then $F_Y(y) = 1 - F_X(g^{-1}(y))$ for $y \in \mathcal{Y}$.*

Proof. A monotone g is one-to-one from $\mathcal{X}$ onto $\mathcal{Y}$, so g^{-1} is single-valued and $g^{-1}(y) = x \iff y = g(x)$. If g is increasing, applying the increasing map g^{-1} to both sides of $g(x) \leq y$ preserves the inequality, so

$$\{x \in \mathcal{X} : g(x) \leq y\} = \{x \in \mathcal{X} : x \leq g^{-1}(y)\},$$

and (33) becomes $\int_{-\infty}^{g^{-1}(y)} f_X(x) dx = F_X(g^{-1}(y))$. If g is decreasing then g^{-1} is decreasing, the inequality reverses, $\{x : g(x) \leq y\} = \{x : x \geq g^{-1}(y)\}$, and the integral is $\int_{g^{-1}(y)}^{\infty} f_X = 1 - F_X(g^{-1}(y))$. $\square$

Theorem 2.22 (Change of variables). *Let X have pdf f_X continuous on $\mathcal{X}$, let $Y = g(X)$ with g monotone, and suppose g^{-1} has a continuous derivative on $\mathcal{Y}$. Then*

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| & y \in \mathcal{Y}, \\ 0 & \text{otherwise.} \end{cases} \quad (34)$$

Proof. Differentiate Theorem 2.21 by the chain rule:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) & \text{if } g \text{ is increasing,} \\ -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) & \text{if } g \text{ is decreasing.} \end{cases}$$

In the first case g^{-1} is increasing and its derivative is positive; in the second it is decreasing and its derivative is negative. So in both cases the coefficient equals $\left| \frac{d}{dy} g^{-1}(y) \right|$. $\square$

Most transformations of interest are not monotone, but are monotone *piecewise*, and that is enough: partition the support and add the contributions.

Theorem 2.23 (Piecewise monotone transformations). *Let X have pdf f_X and $Y = g(X)$. Suppose $\mathcal{X}$ is partitioned into $A_0, A_1, \dots, A_k$ with $\mathbb{P}(X \in A_0) = 0$, and that there are functions $g_1, \dots, g_k$ with $g = g_i$ on A_i , each g_i monotone on A_i with $g_i(A_i) = \mathcal{Y}$ for every i , and each g_i^{-1} continuously differentiable on $\mathcal{Y}$. Then*

$$f_Y(y) = \sum_{i=1}^k f_X(g_i^{-1}(y)) \left| \frac{d}{dy} g_i^{-1}(y) \right|, \quad y \in \mathcal{Y}, \quad (35)$$

and $f_Y(y) = 0$ otherwise.

The exceptional set A_0 is a technical device for endpoints; it is discarded because it carries no probability. Each g_i is a bijection from A_i onto $\mathcal{Y}$, so (35) is (34) applied on each piece and summed.

Example 2.24 (Normal to chi squared). Let X have the standard normal density $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ on $\mathbb{R}$, and let $Y = X^2$. The map $g(x) = x^2$ is decreasing on $(-\infty, 0)$ and increasing on $(0, \infty)$, so take

$$A_0 = \{0\}, \quad A_1 = (-\infty, 0) \text{ with } g_1^{-1}(y) = -\sqrt{y}, \quad A_2 = (0, \infty) \text{ with } g_2^{-1}(y) = \sqrt{y},$$

and $\mathcal{Y} = (0, \infty)$. Since $\left| \frac{d}{dy} (\pm\sqrt{y}) \right| = \frac{1}{2\sqrt{y}}$, (35) gives

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y/2} \frac{1}{2\sqrt{y}} + \frac{1}{\sqrt{2\pi}} e^{-y/2} \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{y}} e^{-y/2}, \quad 0 < y < \infty,$$

the chi squared density with one degree of freedom.

Theorem 2.25 (Probability integral transformation). *Let X have a continuous cdf F_X and set $Y = F_X(X)$. Then Y is uniformly distributed on $(0, 1)$.*

Proof. If F_X is flat on an interval, F_X^{-1} is not defined by inversion, so use the generalized inverse

$$F_X^{-1}(y) = \inf\{x : F_X(x) \geq y\}, \quad 0 < y < 1, \quad (36)$$

which agrees with the ordinary inverse when F_X is strictly increasing. For $0 < y < 1$,

$$\mathbb{P}(Y \leq y) = \mathbb{P}(F_X(X) \leq y) = \mathbb{P}(X \leq F_X^{-1}(y)) = F_X(F_X^{-1}(y)) = y.$$

The middle equality needs a word. If F_X is strictly increasing it is immediate. If F_X is flat on $[x_1, x_2]$ then $F_X^{-1}(F_X(x)) = x_1$ for every x in that interval, so the two events differ; but a flat stretch is a region of zero probability, $\mathbb{P}(x_1 < X \leq x) = F_X(x) - F_X(x_1) = 0$, so the two events have the same probability, which is all that is required. $\square$

3 Lecture 3: Expectation, Moments, and Generating Functions

3.1 Expected values

Definition 3.1 (Expected value). The **expected value** or **mean** of $g(X)$ is

$$\mathbb{E}[g(X)] = \begin{cases} \int_{-\infty}^{\infty} g(x) f_X(x) dx & X \text{ continuous,} \\ \sum_{x \in \mathcal{X}} g(x) f_X(x) & X \text{ discrete,} \end{cases} \quad (37)$$

provided the integral or sum exists. If $\mathbb{E}[|g(X)|] = \infty$ we say that $\mathbb{E}[g(X)]$ **does not exist**.

If X has the Cauchy density $f_X(x) = \pi^{-1}(1+x^2)^{-1}$ on $\mathbb{R}$, which integrates to one, then

$$\mathbb{E}[|X|] = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{\pi} \lim_{M \rightarrow \infty} \log(1+M^2) = \infty, \quad (38)$$

so $\mathbb{E}[X]$ does not exist.

Theorem 3.2 (Expectation is linear and monotone). *Let a, b, c be constants and g_1, g_2 functions whose expectations exist. Then*

- (a) $\mathbb{E}[ag_1(X) + bg_2(X) + c] = a\mathbb{E}[g_1(X)] + b\mathbb{E}[g_2(X)] + c$;
- (b) if $g_1 \geq 0$ everywhere, then $\mathbb{E}[g_1(X)] \geq 0$;
- (c) if $g_1 \geq g_2$ everywhere, then $\mathbb{E}[g_1(X)] \geq \mathbb{E}[g_2(X)]$;
- (d) if $a \leq g_1 \leq b$ everywhere, then $a \leq \mathbb{E}[g_1(X)] \leq b$.

Proof. All four are inherited from the corresponding properties of the integral (or sum). For (a) in the continuous case, additivity of the integral and the fact that constants factor out give

$$\int (ag_1 + bg_2 + c) f_X = a \int g_1 f_X + b \int g_2 f_X + c \int f_X,$$

and $\int f_X = 1$ by Theorem 2.20. Part (b) is monotonicity of the integral of a nonnegative function; (c) follows by applying (b) to $g_1 - g_2$ and then (a); (d) follows from (c) with the constant functions a and b . $\square$

Example 3.3 (The mean is the best constant predictor). Measure the distance between X and a constant b by $\mathbb{E}[(X - b)^2]$, and ask which b minimizes it. Adding and subtracting $\mathbb{E}[X]$ and expanding,

$$\begin{aligned}\mathbb{E}[(X - b)^2] &= \mathbb{E}[(X - \mathbb{E}[X]) + (\mathbb{E}[X] - b)^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X])^2] + (\mathbb{E}[X] - b)^2 + 2(\mathbb{E}[X] - b)\mathbb{E}[X - \mathbb{E}[X]],\end{aligned}\quad (39)$$

where $(\mathbb{E}[X] - b)$ came out of the expectation as a constant. The last term vanishes because $\mathbb{E}[X - \mathbb{E}[X]] = \mathbb{E}[X] - \mathbb{E}[X] = 0$, leaving

$$\mathbb{E}[(X - b)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2] + (\mathbb{E}[X] - b)^2. \quad (40)$$

The first term does not involve b and the second is nonnegative and vanishes at $b = \mathbb{E}[X]$. Hence $\min_b \mathbb{E}[(X - b)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2]$, attained at $b = \mathbb{E}[X]$.

3.2 Moments and moment generating functions

Definition 3.4 (Moments, variance). For an integer $n \geq 1$, the n th **moment** of X is $\mu'_n = \mathbb{E}[X^n]$, and the n th **central moment** is $\mu_n = \mathbb{E}[(X - \mu)^n]$ where $\mu = \mathbb{E}[X]$. The second central moment is the **variance**, $\text{Var } X = \mathbb{E}[(X - \mathbb{E}[X])^2]$, and its positive square root is the **standard deviation**.

Two computational facts. First, expanding the square and using linearity,

$$\begin{aligned}\text{Var } X &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,\end{aligned}\quad (41)$$

usually the easier route. Second, variance is invariant to shifts and scales quadratically: since $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$,

$$\text{Var}(aX + b) = \mathbb{E}[(aX + b - a\mathbb{E}[X] - b)^2] = a^2 \mathbb{E}[(X - \mathbb{E}[X])^2] = a^2 \text{Var } X. \quad (42)$$

Example 3.5 (The normal family). Let Z have the standard normal density

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in \mathbb{R}, \quad (43)$$

which integrates to one by the Gaussian integral (Proposition 4.5). Symmetry gives $\mathbb{E}[Z] = 0$, and integrating by parts with $u = z$, $dv = z\varphi(z) dz = -d\varphi(z)$,

$$\text{Var } Z = \mathbb{E}[Z^2] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z \cdot z e^{-z^2/2} dz = \underbrace{[-z\varphi(z)]_{-\infty}^{\infty}}_{=0} + \int_{-\infty}^{\infty} \varphi(z) dz = 1. \quad (44)$$

Now put $X = \mu + \sigma Z$ with $\sigma > 0$. The map is increasing with inverse $z = (x - \mu)/\sigma$ and Jacobian $1/\sigma$, so Theorem 2.22 gives

$$f_X(x) = \varphi\left(\frac{x - \mu}{\sigma}\right) \frac{1}{\sigma} = \frac{1}{\sqrt{2\pi} \sigma} e^{-(x - \mu)^2/(2\sigma^2)}, \quad x \in \mathbb{R}, \quad (45)$$

the $N(\mu, \sigma^2)$ density, and Theorem 3.2(a) with (42) gives

$$\mathbb{E}[X] = \mu + \sigma \mathbb{E}[Z] = \mu, \quad \text{Var } X = \sigma^2 \text{Var } Z = \sigma^2.$$

Read backwards, $Z = (X - \mu)/\sigma$: every normal probability reduces to one about Z , which is why a single table suffices. The two parameters are exactly the mean and the variance, and (45) is the leading case of the location–scale construction of Section 4.5.

Theorem 3.6 (Jensen’s inequality). *If ϕ is convex and $\mathbb{E}[X]$ exists, then*

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)], \quad (46)$$

with the inequality reversed for ϕ concave, and strict unless ϕ is linear on the support of X .

Proof. Convexity gives a supporting line at $\mu = \mathbb{E}[X]$: there is c with $\phi(x) \geq \phi(\mu) + c(x - \mu)$ for all x ($c = \phi'(\mu)$ when ϕ is differentiable). Taking expectations and using Theorem 3.2(a) and (c),

$$\mathbb{E}[\phi(X)] \geq \phi(\mu) + c(\mathbb{E}[X] - \mu) = \phi(\mu).$$

□

Definition 3.7 (Moment generating function). The **moment generating function** (**mgf**) of X is

$$M_X(t) = \mathbb{E}[e^{tX}], \quad (47)$$

provided the expectation is finite for all t in some neighbourhood $(-h, h)$ of 0. If no such $h > 0$ exists, the mgf does not exist.

Theorem 3.8 (The mgf generates moments). *If X has mgf M_X , then $\mathbb{E}[X^n] = M_X^{(n)}(0)$ for every $n \geq 1$.*

Proof. Differentiating under the integral sign, which is legitimate here by Corollary C.4,

$$\begin{aligned} \frac{d}{dt} M_X(t) &= \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} f_X(x) dx \\ &= \int_{-\infty}^{\infty} \frac{\partial}{\partial t} e^{tx} f_X(x) dx = \int_{-\infty}^{\infty} x e^{tx} f_X(x) dx = \mathbb{E}[X e^{tX}]. \end{aligned} \quad (48)$$

Evaluating at $t = 0$ gives $\mathbb{E}[X]$. Iterating n times gives $M_X^{(n)}(t) = \mathbb{E}[X^n e^{tX}]$, and at $t = 0$ this is $\mathbb{E}[X^n]$. $\square$

Example 3.9 (Moments need not determine a distribution). Let

$$f_1(x) = \frac{1}{\sqrt{2\pi}x} e^{-(\log x)^2/2}, \quad f_2(x) = f_1(x) [1 + \sin(2\pi \log x)], \quad 0 < x < \infty,$$

the first being the lognormal density. If $X_1 \sim f_1$ then $\mathbb{E}[X_1^r] = e^{r^2/2}$ for every r . If $X_2 \sim f_2$ then

$$\mathbb{E}[X_2^r] = \mathbb{E}[X_1^r] + \int_0^\infty x^r f_1(x) \sin(2\pi \log x) dx, \quad (49)$$

and the substitution $y = \log x - r$ turns the last integral into that of an odd function over $\mathbb{R}$, hence zero, for every $r = 0, 1, 2, \dots$. So X_1 and X_2 have different densities and identical moments of every order. Note that neither has an mgf: the lognormal has all moments finite but $\mathbb{E}[e^{tX}] = \infty$ for every $t > 0$.

Theorem 3.10 (Characterization). *Let F_X and F_Y be cdfs all of whose moments exist.*

- (a) If X and Y have bounded support, then $F_X = F_Y$ if and only if $\mathbb{E}[X^r] = \mathbb{E}[Y^r]$ for all integers $r \geq 0$.
- (b) If the mgfs exist and $M_X(t) = M_Y(t)$ for all t in some neighbourhood of 0, then $F_X = F_Y$.

Theorem 3.11 (Convergence of mgfs). *Let $X_1, X_2, \dots$ have mgfs M_{X_i} , and suppose $M_{X_i}(t) \rightarrow M_X(t)$ for all t in a neighbourhood of 0, where M_X is itself an mgf, of a cdf F_X . Then $F_{X_i}(x) \rightarrow F_X(x)$ at every continuity point x of F_X .*

Proof omitted; see Billingsley [1995, §30].

Theorem 3.12 (Affine transformations). *For constants a and b ,*

$$M_{aX+b}(t) = e^{bt} M_X(at).$$

Proof. Directly from (47) and Theorem 3.2(a),

$$M_{aX+b}(t) = \mathbb{E}[e^{(aX+b)t}] = \mathbb{E}[e^{(at)X} e^{bt}] = e^{bt} \mathbb{E}[e^{(at)X}] = e^{bt} M_X(at).$$

□

4 Lecture 4: Common Families of Distributions

4.1 Three basic families

4.1.1 Bernoulli

Definition 4.1 (Bernoulli). $X \sim \text{Bernoulli}(p)$, $p \in [0, 1]$, if

$$f(x | p) = \begin{cases} p & x = 1, \\ 1 - p & x = 0, \\ 0 & \text{otherwise.} \end{cases} \quad (50)$$

Proposition 4.2. $\mathbb{E}[X] = p$, $\text{Var } X = p(1 - p)$, $M_X(t) = 1 - p + pe^t$.

Proof. Directly from (37), the sum having two terms:

$$\begin{aligned} \mathbb{E}[X] &= 0 \cdot (1 - p) + 1 \cdot p = p, \\ \text{Var } X &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p), \end{aligned} \quad (51)$$

and $M_X(t) = e^0(1 - p) + e^t p = 1 - p + pe^t$. □

4.1.2 Uniform

Definition 4.3 (Uniform). $X \sim \text{U}\{1, \dots, N\}$ if $f(x) = 1/N$ for $x = 1, \dots, N$; $X \sim \text{U}(a, b)$ if

$$f(x | a, b) = \frac{1}{b - a} \mathbf{1}\{a < x < b\}. \quad (52)$$

Proposition 4.4. For $X \sim \text{U}(a, b)$, $\mathbb{E}[X] = \frac{a + b}{2}$ and $\text{Var } X = \frac{(b - a)^2}{12}$.

Proof. Both moments are elementary integrals:

$$\begin{aligned} \mathbb{E}[X] &= \frac{1}{b - a} \int_a^b x \, dx = \frac{b^2 - a^2}{2(b - a)} = \frac{a + b}{2}, \\ \mathbb{E}[X^2] &= \frac{1}{b - a} \int_a^b x^2 \, dx = \frac{b^3 - a^3}{3(b - a)} = \frac{a^2 + ab + b^2}{3}, \end{aligned} \quad (53)$$

and (41) gives

$$\text{Var } X = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}.$$

□

4.1.3 Gaussian

Proposition 4.5 (Gaussian integral).

$$\int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}. \quad (54)$$

Proof. See the footnote.²

□

Definition 4.6 (Normal). $X \sim N(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$, if

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, \quad x \in \mathbb{R}, \quad (55)$$

which is a density by Proposition 4.5 after the substitution $z = (x - \mu)/\sigma$.

Proposition 4.7. $M_X(t) = e^{\mu t + \sigma^2 t^2/2}$, and hence $\mathbb{E}[X] = \mu$, $\text{Var } X = \sigma^2$.

Proof. Take $Z \sim N(0, 1)$ first and complete the square in the exponent:

$$M_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz - z^2/2} dz = e^{t^2/2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-t)^2/2} dz = e^{t^2/2}, \quad (56)$$

because $tz - \frac{z^2}{2} = -\frac{(z-t)^2}{2} + \frac{t^2}{2}$ and the remaining integrand is the $N(t, 1)$ density. Since $X = \mu + \sigma Z$ (Example 3.5), Theorem 3.12 gives $M_X(t) = e^{\mu t} M_Z(\sigma t) = e^{\mu t + \sigma^2 t^2/2}$.

²The Gaussian integral. Put $I = \int_{-\infty}^{\infty} e^{-z^2/2} dz$. The integrand has no elementary antiderivative, so square it and read the result as a double integral over the plane, which Tonelli's theorem licenses because the integrand is nonnegative:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2} dy \right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dx dy.$$

Now change to polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, whose Jacobian is r , so that $dx dy = r dr d\theta$ and $x^2 + y^2 = r^2$:

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = \int_0^{2\pi} \left[-e^{-r^2/2} \right]_0^{\infty} d\theta = \int_0^{2\pi} 1 d\theta = 2\pi.$$

The extra factor r supplied by the Jacobian is exactly what makes the radial integral elementary. Hence $I = \sqrt{2\pi}$, taking the positive root because the integrand is positive. Equivalently $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, which is $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ in (104).

The cumulant function $K_X(t) = \log M_X(t) = \mu t + \sigma^2 t^2/2$ is quadratic, so $K'_X(0) = \mu$ and $K''_X(0) = \sigma^2$, and every cumulant beyond the second vanishes. $\square$

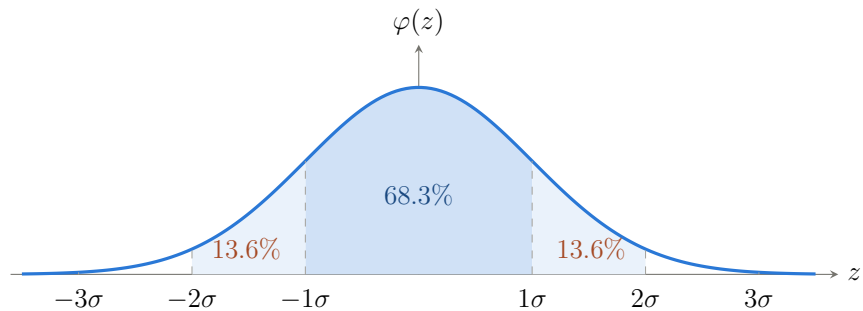


Figure 5: The standard normal density $\varphi(z) = (2\pi)^{-1/2}e^{-z^2/2}$, whose normalizing constant is Proposition 4.5. Inflection points sit at $\pm\sigma$, and the shaded bands carry 68.3% and 95.4% of the mass.

4.2 Discrete families

Table 2: Discrete families: density, mean, and variance.

Family	pmf $f(x)$ and support	$\mathbb{E}[X]$	$\text{Var } X$
Uniform $\{1, \dots, N\}$	$\frac{1}{N}, x = 1, \dots, N$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$
Bernoulli (p)	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$
Binomial (n, p)	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, \dots, n$	np	$np(1-p)$
Poisson (λ)	$\frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, \dots$	λ	λ
Geometric (p)	$(1-p)^{x-1} p, x = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative binomial (r, p)	$\binom{x-1}{r-1} p^r (1-p)^{x-r},$ $x = r, r+1, \dots$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Hypergeometric (N, K, n)	$\frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, x \text{ feasible}$	$\frac{nK}{N}$	$\frac{nK}{N} \frac{N-K}{N} \frac{N-n}{N-1}$

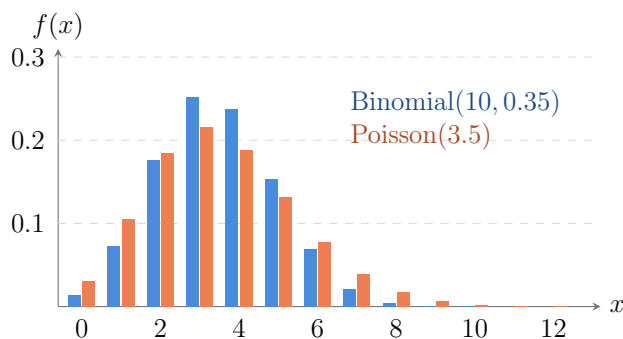
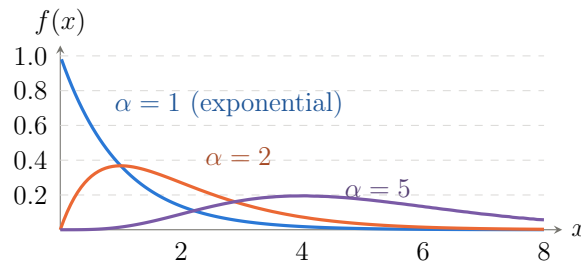


Figure 6: Two discrete families with the same mean 3.5. The binomial has variance $np(1-p) = 2.275 < 3.5$; the Poisson has variance equal to its mean, so it puts more weight in the right tail and, unlike the binomial, has unbounded support.

4.3 Continuous families

Table 3: Continuous families: density, mean, and variance.

Family	pdf $f(x)$ and support	$\mathbb{E}[X]$	$\text{Var } X$
Uniform (a, b)	$\frac{1}{b-a}, \quad a < x < b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential (β)	$\frac{1}{\beta}e^{-x/\beta}, \quad x > 0$	β	β^2
Gamma (α, β)	$\frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}, \quad x > 0$	$\alpha\beta$	$\alpha\beta^2$
Chi squared (p)	$\frac{x^{p/2-1}e^{-x/2}}{\Gamma(p/2)2^{p/2}}, \quad x > 0$	p	$2p$
Beta (α, β)	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{\text{B}(\alpha, \beta)}, \quad 0 < x < 1$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Normal (μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/(2\sigma^2)}, \quad x \in \mathbb{R}$	μ	σ^2
Lognormal (μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma x}e^{-(\log x - \mu)^2/(2\sigma^2)}, \quad x > 0$	$e^{\mu+\sigma^2/2}$	$e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$
Double exponential (μ, σ)	$\frac{1}{2\sigma}e^{- x-\mu /\sigma}, \quad x \in \mathbb{R}$	μ	$2\sigma^2$
Cauchy (θ, σ)	$\frac{1}{\pi\sigma} \frac{1}{1+((x-\theta)/\sigma)^2}, \quad x \in \mathbb{R}$	does not exist	does not exist
Weibull (γ, β)	$\frac{\gamma}{\beta}x^{\gamma-1}e^{-x^\gamma/\beta}, \quad x > 0$	$\beta^{1/\gamma}\Gamma\left(1+\frac{1}{\gamma}\right)$	$\beta^{2/\gamma}\left[\Gamma\left(1+\frac{2}{\gamma}\right)-\Gamma\left(1+\frac{1}{\gamma}\right)^2\right]$

**Figure 7:** Gamma densities with $\beta = 1$. The mean $\alpha\beta$ moves right and the shape passes from a decreasing curve at $\alpha = 1$ to a mode at $(\alpha - 1)\beta$; by the central limit theorem the shape approaches the normal as $\alpha \rightarrow \infty$.

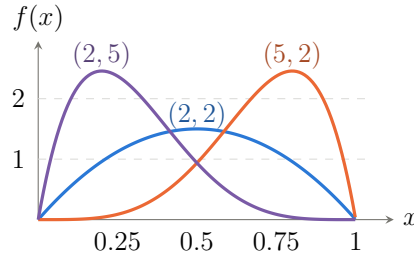


Figure 8: Beta densities on $(0, 1)$, labelled (α, β) . The mean $\alpha/(\alpha + \beta)$ shifts toward whichever parameter is larger, and $(1, 1)$ is the uniform. Its support makes it the standard prior for a probability.

4.4 Limits and characterizations

Example 4.8 (Poisson limit of the binomial). Let $X_n \sim \text{Binomial}(n, p_n)$ with $np_n \rightarrow \lambda$. By Table 2 and Theorem 3.11,

$$M_{X_n}(t) = (1 + p_n(e^t - 1))^n = \left(1 + \frac{np_n(e^t - 1)}{n}\right)^n \rightarrow e^{\lambda(e^t - 1)}, \quad (57)$$

using $(1 + c_n/n)^n \rightarrow e^c$ whenever $c_n \rightarrow c$. The limit is the Poisson mgf, so $X_n \rightarrow \text{Poisson}(\lambda)$ in distribution.

Proposition 4.9 (Memorylessness of the geometric). *If $X \sim \text{Geometric}(p)$ then $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$ for integers $s, t \geq 0$, and the geometric is the only discrete family with this property.*

Proof. $\mathbb{P}(X > k) = (1 - p)^k$, since exceeding k means the first k trials all failed. Hence

$$\mathbb{P}(X > s + t \mid X > s) = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} = \frac{(1 - p)^{s+t}}{(1 - p)^s} = (1 - p)^t = \mathbb{P}(X > t).$$

Conversely $G(k) = \mathbb{P}(X > k)$ satisfying $G(s + t) = G(s)G(t)$ with $G(1) = 1 - p$ forces $G(k) = (1 - p)^k$ by induction. $\square$

Proposition 4.10 (Memorylessness of the exponential). *$X \sim \text{Exponential}(\beta)$ satisfies $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$ for all $s, t > 0$, and is the only continuous family that does.*

Proof. $\mathbb{P}(X > x) = \int_x^\infty \beta^{-1} e^{-u/\beta} du = e^{-x/\beta}$, so the ratio $e^{-(s+t)/\beta} / e^{-s/\beta} = e^{-t/\beta}$ is $\mathbb{P}(X > t)$. Conversely, $G(x) = \mathbb{P}(X > x)$ is right-continuous, decreasing, and satisfies

the Cauchy functional equation $G(s+t) = G(s)G(t)$, whose only such solutions are $G(x) = e^{-cx}$ with $c > 0$. $\square$

4.5 Location and scale families

Definition 4.11 (Location–scale family). Let f be a pdf. The family generated by f is

$$f(x \mid \mu, \sigma) = \frac{1}{\sigma} f\left(\frac{x - \mu}{\sigma}\right), \quad \mu \in \mathbb{R}, \sigma > 0. \quad (58)$$

Setting $\sigma = 1$ gives a **location** family; $\mu = 0$ a **scale** family.

Theorem 4.12. $X \sim f(x \mid \mu, \sigma)$ if and only if $X = \mu + \sigma Z$ with $Z \sim f(z)$. Consequently $\mathbb{E}[X] = \mu + \sigma \mathbb{E}[Z]$ and $\text{Var } X = \sigma^2 \text{Var } Z$ whenever these exist.

Proof. ($\Leftarrow$) Let $Z \sim f$ and put $X = \mu + \sigma Z$. The map $g(z) = \mu + \sigma z$ is strictly increasing on $\mathbb{R}$, since $\sigma > 0$, with inverse $g^{-1}(x) = (x - \mu)/\sigma$ and $\frac{d}{dx}g^{-1}(x) = 1/\sigma > 0$. By Theorem 2.22,

$$f_X(x) = f(g^{-1}(x)) \left| \frac{d}{dx}g^{-1}(x) \right| = \frac{1}{\sigma} f\left(\frac{x - \mu}{\sigma}\right),$$

which is (58).

($\Rightarrow$) Let $X \sim f(x \mid \mu, \sigma)$ and set $Z = (X - \mu)/\sigma$, the image of X under the strictly increasing map g^{-1} . Applying Theorem 2.22 to g^{-1} , whose inverse is g with derivative σ ,

$$f_Z(z) = f_X(\mu + \sigma z) \sigma = \frac{1}{\sigma} f\left(\frac{\mu + \sigma z - \mu}{\sigma}\right) \sigma = f(z),$$

so $Z \sim f$ and $X = \mu + \sigma Z$ as required. The two directions are inverse to each other because g is a bijection of $\mathbb{R}$.

For the moments, $X = \mu + \sigma Z$ is affine in Z , so Theorem 3.2(a) gives $\mathbb{E}[X] = \mu + \sigma \mathbb{E}[Z]$ and (42) gives $\text{Var } X = \sigma^2 \text{Var } Z$, both whenever the right-hand sides exist. Note that μ and σ are the mean and standard deviation only when $\mathbb{E}[Z] = 0$ and $\text{Var } Z = 1$; in general they are location and scale parameters, nothing more, as the Cauchy shows. $\square$

Normal (Example 3.5), Cauchy, double exponential and logistic (Example 2.10) are location–scale families; exponential and gamma with fixed α are scale families.

5 Lecture 5: Exponential Families and Random Vectors

5.1 Exponential families

Definition 5.1 (Exponential family). A family of densities $\{f(x \mid \theta) : \theta \in \Theta\}$ is an **exponential family** if it can be written as

$$f(x \mid \theta) = \underbrace{h(x)}_{\text{carrier}} \underbrace{c(\theta)}_{\text{normalizer}} \exp \left(\sum_{i=1}^k \underbrace{w_i(\theta)}_{\text{parameter}} \underbrace{t_i(x)}_{\text{statistic}} \right) \quad (59)$$

for some k , where

$$h : \mathbb{R} \rightarrow \mathbb{R}_+, \quad c : \Theta \rightarrow \mathbb{R}_{++}, \quad t_i : \mathbb{R} \rightarrow \mathbb{R}, \quad w_i : \Theta \rightarrow \mathbb{R}, \quad (60)$$

so that each t_i is a function of x alone and each w_i a function of θ alone. The support $\{x : f(x \mid \theta) > 0\} = \{x : h(x) > 0\}$ is therefore free of θ .

The content of (59) is a *separation*: the only place where the data and the parameter meet is the sum $\sum_i w_i(\theta) t_i(x)$, and there they meet through the k numbers $t_1(x), \dots, t_k(x)$ and nothing else. Reading the four ingredients:

Collecting the two lists into vectors,

$$\mathbf{t}(x) = (t_1(x), \dots, t_k(x))^\top, \quad \mathbf{w}(\theta) = (w_1(\theta), \dots, w_k(\theta))^\top,$$

the four ingredients are these.

- $\mathbf{t}(x)$ is the vector of **sufficient statistics**.
- $\mathbf{w}(\theta)$ is the vector of **natural parameters**, a reparametrization of θ .
- $h(x) \geq 0$ is the **carrier**, the part of the density that does not involve θ at all.

- $c(\theta) > 0$ is the **normalizer**, fixed by the requirement that f integrate to one:

$$c(\theta) = \begin{cases} \left[\sum_{x \in \mathbb{R}} h(x) \exp \left\{ \sum_{j=1}^k w_j(\theta) t_j(x) \right\} \right]^{-1}, & X \text{ discrete,} \\ \left[\int_{\mathbb{R}} h(x) \exp \left\{ \sum_{j=1}^k w_j(\theta) t_j(x) \right\} dx \right]^{-1}, & X \text{ continuous.} \end{cases} \quad (61)$$

Since $\mathbf{w}$ merely relabels the parameter, we may as well use it as the parameter. Writing $\eta_i = w_i(\theta)$ and absorbing c into the exponent through $c(\theta) = e^{-A(\boldsymbol{\eta})}$, (59) becomes the **natural** (or canonical) form

$$f(x \mid \boldsymbol{\eta}) = h(x) \exp(\boldsymbol{\eta}^\top \mathbf{t}(x) - A(\boldsymbol{\eta})), \quad A(\boldsymbol{\eta}) = \log \int h(x) e^{\boldsymbol{\eta}^\top \mathbf{t}(x)} dx, \quad (62)$$

where A is the **log-partition** or **cumulant** function. Comparing with (61), $A(\boldsymbol{\eta}) = -\log c(\theta)$: it is the logarithm of the reciprocal of the same normalizing constant, with the sum replacing the integral in the discrete case. The family is the set of $\boldsymbol{\eta}$ for which that integral is finite.

Theorem 5.2 (Moments from the log-partition function). *For the natural form (62),*

$$\mathbb{E}[\mathbf{t}(X)] = \nabla A(\boldsymbol{\eta}), \quad \text{Var}(\mathbf{t}(X)) = \nabla^2 A(\boldsymbol{\eta}). \quad (63)$$

Proof. Write

$$Z(\boldsymbol{\eta}) = \int h(x) e^{\boldsymbol{\eta}^\top \mathbf{t}(x)} dx, \quad \text{so that} \quad A = \log Z, \quad f(x \mid \boldsymbol{\eta}) = \frac{h(x) e^{\boldsymbol{\eta}^\top \mathbf{t}(x)}}{Z(\boldsymbol{\eta})}.$$

Differentiating under the integral sign, legitimate by Corollary C.4,

$$\frac{\partial Z}{\partial \eta_i} = \int t_i(x) h(x) e^{\boldsymbol{\eta}^\top \mathbf{t}(x)} dx,$$

and therefore

$$\frac{\partial A}{\partial \eta_i} = \frac{1}{Z} \frac{\partial Z}{\partial \eta_i} = \int t_i(x) f(x \mid \boldsymbol{\eta}) dx = \mathbb{E}[t_i(X)],$$

which is the first half of (63). Differentiating that identity once more in η_j , and using $\partial f / \partial \eta_j = (t_j(x) - \mathbb{E}[t_j(X)]) f(x | \boldsymbol{\eta})$, which follows from the previous display,

$$\begin{aligned} \frac{\partial^2 A}{\partial \eta_i \partial \eta_j} &= \int t_i(x) (t_j(x) - \mathbb{E}[t_j(X)]) f(x | \boldsymbol{\eta}) dx \\ &= \mathbb{E}[t_i(X) t_j(X)] - \mathbb{E}[t_i(X)] \mathbb{E}[t_j(X)] = \text{Cov}(t_i(X), t_j(X)). \end{aligned}$$

That is the (i, j) entry of $\text{Var}(\mathbf{t}(X))$. Incidentally the Hessian of A is a covariance matrix, hence positive semidefinite, so A is convex. $\square$

Example 5.3 (Putting the Poisson in the form). Start from the pmf and force the parameter into an exponent:

$$f(x | \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} = \underbrace{\frac{1}{x!}}_{h(x)} \underbrace{e^{-\lambda}}_{c(\lambda)} \exp\left(\underbrace{(\log \lambda)}_{w_1(\lambda)} \underbrace{x}_{t_1(x)}\right),$$

using $\lambda^x = e^{x \log \lambda}$. So $k = 1$, the sufficient statistic is $t(x) = x$, and the natural parameter is $\eta = \log \lambda$. Inverting, $\lambda = e^\eta$, and

$$A(\eta) = \log \sum_{x \geq 0} \frac{e^{\eta x}}{x!} = \log e^{e^\eta} = e^\eta = \lambda,$$

the exponential series, which is (61) in the discrete case read through $A = -\log c$. Theorem 5.2 then delivers both moments with two derivatives and no integration: $A'(\eta) = e^\eta = \lambda = \mathbb{E}[X]$ and $A''(\eta) = e^\eta = \lambda = \text{Var } X$, which is the equality of mean and variance recorded in Table 2.

Example 5.4 (Bernoulli). Let $X \sim \text{Bernoulli}(p)$ with $p \in (0, 1)$, so that

$$f(x | p) = \begin{cases} p & x = 1, \\ 1 - p & x = 0, \\ 0 & \text{otherwise,} \end{cases} \quad \text{that is} \quad f(x | p) = p^x (1 - p)^{1-x} \mathbf{1}\{x \in \{0, 1\}\}.$$

Write the two factors as exponentials and collect the terms in x :

$$p^x (1 - p)^{1-x} = \exp\{x \log p + (1 - x) \log(1 - p)\} = (1 - p) \exp\left\{x \log \frac{p}{1 - p}\right\},$$

so that

$$f(x | p) = \underbrace{\mathbf{1}\{x \in \{0, 1\}\}}_{h(x)} \underbrace{(1-p)}_{c(p)} \exp\left(\underbrace{\log \frac{p}{1-p}}_{w_1(p)} \underbrace{x}_{t_1(x)}\right).$$

Again $k = 1$ and $t(x) = x$. The support is $\{x : h(x) > 0\} = \{0, 1\}$, free of p as Definition 5.1 requires. The natural parameter is the **logit** $\eta = \log \frac{p}{1-p}$, which inverts to $p = e^\eta / (1 + e^\eta)$, and (61) gives the normalizer as a two-term sum:

$$c(p)^{-1} = \sum_{x \in \{0,1\}} e^{\eta x} = 1 + e^\eta, \quad \text{so} \quad A(\eta) = \log(1 + e^\eta).$$

Differentiating once and twice, Theorem 5.2 returns

$$A'(\eta) = \frac{e^\eta}{1 + e^\eta} = p = \mathbb{E}[X], \quad A''(\eta) = \frac{e^\eta}{(1 + e^\eta)^2} = p(1-p) = \text{Var } X,$$

reproducing Proposition 4.2 with no summation at all. Note where $p \in (0, 1)$ was needed: at $p = 0$ or $p = 1$ the logit is undefined, and the degenerate distribution sitting at a single point is not a member of the family.

Example 5.5 (A two-dimensional statistic). For the $N(\mu, \sigma^2)$ family, expand the square in the exponent:

$$\begin{aligned} f(x | \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right) \\ &= \underbrace{1}_{h(x)} \underbrace{\frac{1}{\sqrt{2\pi}\sigma} e^{-\mu^2/(2\sigma^2)}}_{c(\mu, \sigma^2)} \exp\left(\underbrace{\frac{\mu}{\sigma^2}}_{w_1(\theta)} \cdot \underbrace{x}_{t_1(x)} + \underbrace{\left(-\frac{1}{2\sigma^2}\right)}_{w_2(\theta)} \cdot \underbrace{x^2}_{t_2(x)}\right), \end{aligned}$$

with $h(x) = 1$: everything else depends on $\theta = (\mu, \sigma^2)$, and the factor $e^{-\mu^2/(2\sigma^2)}$ is the part of the normalizer that moves with μ . Here $k = 2$, and the data enter only through $\mathbf{t}(x) = (x, x^2)^\top$. The natural parameter is $\boldsymbol{\eta} = (\mu/\sigma^2, -1/(2\sigma^2))^\top$, and solving back, $\sigma^2 = -1/(2\eta_2)$ and $\mu = -\eta_1/(2\eta_2)$, so

$$A(\boldsymbol{\eta}) = -\log c(\mu, \sigma^2) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2) = -\frac{\eta_1^2}{4\eta_2} + \frac{1}{2} \log\left(-\frac{\pi}{\eta_2}\right),$$

defined on $\eta_2 < 0$. Differentiating, $\partial A / \partial \eta_1 = -\eta_1 / (2\eta_2) = \mu$ and $\partial^2 A / \partial \eta_1^2 = -1 / (2\eta_2) = \sigma^2$, again the mean and the variance.

Table 4: Exponential-family representation. The carrier h collects what does not involve the parameter; the sufficient statistic $\mathbf{t}$ is the only channel through which the data reach the parameter; $\boldsymbol{\eta}$ is the natural parameter of (62); and A is the log-partition function, whose derivatives are the moments of $\mathbf{t}$ by Theorem 5.2. The binomial requires $p \in (0, 1)$, the range on which the logit is defined.

Family	$h(x)$	$\mathbf{t}(x)$	$\boldsymbol{\eta}$	$A(\boldsymbol{\eta})$
Binomial(n, p)	$\binom{n}{x}$	x	$\log \frac{p}{1-p}$	$n \log(1 + e^\eta)$
Poisson(λ)	$1/x!$	x	$\log \lambda$	e^η
Normal(μ, σ^2)	1	(x, x^2)	$(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2})$	$-\frac{\eta_1^2}{4\eta_2} + \frac{1}{2} \log(-\pi/\eta_2)$
Gamma(α, β)	1	$(\log x, x)$	$(\alpha - 1, -\frac{1}{\beta})$	$\log \Gamma(\eta_1 + 1) - (\eta_1 + 1) \log(-\eta_2)$
Beta(α, β)	1	$(\log x, \log(1 - x))$	$(\alpha - 1, \beta - 1)$	$\log B(\eta_1 + 1, \eta_2 + 1)$

The support condition in Definition 5.1 is not a technicality. The $U[0, \theta]$ family is *not* exponential: its support $\{0 < x < \theta\}$ moves with θ , so the indicator $\mathbf{1}\{0 < x < \theta\}$ cannot be split into a factor free of θ times a factor free of x , and no representation (59) exists.

5.2 Random vectors and marginal distributions

Definition 5.6 (Random vector; joint distribution). An n -dimensional **random vector** is a measurable map $\mathbf{X} = (X_1, \dots, X_n)^\top : S \rightarrow \mathbb{R}^n$. Vectors are written in bold and are columns throughout. Its **joint cdf** is $F_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n)$, and its **joint pmf** or **pdf** $f_{\mathbf{X}}$ satisfies, for $A \in \mathcal{B}(\mathbb{R}^n)$,

$$\mathbb{P}(\mathbf{X} \in A) = \sum_{\mathbf{x} \in A} f_{\mathbf{X}}(\mathbf{x}) \quad \text{or} \quad \int_A f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}. \quad (64)$$

For any $g : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbb{E}[g(\mathbf{X})] = \int_{\mathbb{R}^n} g(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}$ (sum in the discrete case).

Take $n = 2$ and write $\mathbf{X} = (X, Y)^\top$. The joint cdf is a function of two arguments, $F_{X,Y} : \mathbb{R}^2 \rightarrow [0, 1]$, and (64) unfolds into a double sum or a double integral:

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \begin{cases} \sum_{s \leq x} \sum_{t \leq y} f_{X,Y}(s, t), & \text{discrete,} \\ \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) dt ds, & \text{continuous.} \end{cases} \quad (65)$$

In the continuous case the density is recovered by differentiating once in each argument, $f_{X,Y}(x, y) = \partial^2 F_{X,Y}(x, y) / \partial x \partial y$. Probabilities of rectangles come from (65) by inclusion and exclusion, as in (8): for $a < b$ and $c < d$,

$$\mathbb{P}(a < X \leq b, c < Y \leq d) = F_{X,Y}(b, d) - F_{X,Y}(a, d) - F_{X,Y}(b, c) + F_{X,Y}(a, c). \quad (66)$$

Letting one argument run off to infinity recovers the cdf of the other coordinate in isolation,

$$F_X(x) = \mathbb{P}(X \leq x) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y), \quad F_Y(y) = \lim_{x \rightarrow \infty} F_{X,Y}(x, y). \quad (67)$$

Definition 5.7 (Marginal distributions). For (X, Y) with joint pmf or pdf $f(x, y)$, the **marginals** are obtained by summing or integrating out the other coordinate:

$$f_X(x) = \begin{cases} \sum_y f(x, y), & \text{discrete,} \\ \int_{-\infty}^{\infty} f(x, y) dy, & \text{continuous,} \end{cases} \quad f_Y(y) = \begin{cases} \sum_x f(x, y), & \text{discrete,} \\ \int_{-\infty}^{\infty} f(x, y) dx, & \text{continuous.} \end{cases} \quad (68)$$

6 Lecture 6: Conditioning and Independence

6.1 Conditional distributions

Definition 6.1 (Conditional pmf and pdf). Let (X, Y) have joint pmf or pdf $f(x, y)$ with marginals (68). The **conditional distribution of X given $Y = y$** is

$$f_{X|Y}(x | y) = \frac{f(x, y)}{f_Y(y)} \quad \text{for every } y \text{ with } f_Y(y) > 0, \quad (69)$$

and symmetrically the **conditional distribution of Y given $X = x$** is

$$f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)} \quad \text{for every } x \text{ with } f_X(x) > 0. \quad (70)$$

In the discrete case these are pmfs in the free argument; in the continuous case they are densities.³

Proposition 6.2 (The discrete case). *If X and Y are discrete and $f_Y(y) = \mathbb{P}(Y = y) > 0$, then*

$$f_{X|Y}(x | y) = \mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}. \quad (71)$$

Proof. Apply Definition 1.16 with $A = \{X = x\}$ and $B = \{Y = y\}$, which is legitimate because $\mathbb{P}(B) = f_Y(y) > 0$. The numerator is $\mathbb{P}(A \cap B) = \mathbb{P}(X = x, Y = y) = f(x, y)$ and the denominator is $f_Y(y)$, so the ratio is (69). $\square$

For each fixed y with $f_Y(y) > 0$, the function $f_{X|Y}(\cdot | y)$ is a genuine pmf or pdf in x : it is nonnegative, and it sums or integrates to $\sum_x f(x, y)/f_Y(y) = 1$ by (68). So the whole apparatus of Section 3 applies to it, one fixed y at a time.

³Equation (69) is a *definition*, in the continuous case $\mathbb{P}(Y = y) = 0$ by Proposition 2.17, so Definition 1.16 cannot produce it. Conditioning on a null event is also ambiguous, the **Borel–Kolmogorov paradox**: for a point uniform on the sphere, conditioning on a meridian leaves the latitude nonuniform along that circle, while conditioning on the equator leaves the longitude uniform, though both are great circles. What is well defined is the random variable $\mathbb{E}[X | Y]$ of (75), tied to the variable Y conditioned on.

Definition 6.3 (Conditional expectation). For $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ and y with $f_Y(y) > 0$,

$$\mathbb{E}[g(X, Y) \mid Y = y] = \begin{cases} \sum_x g(x, y) f_{X|Y}(x \mid y), & \text{discrete,} \\ \int_{-\infty}^{\infty} g(x, y) f_{X|Y}(x \mid y) dx, & \text{continuous,} \end{cases} \quad (72)$$

provided the sum or integral exists. Inside (72) the second argument is frozen at y , so $g(\cdot, y)$ is a function of x alone and (72) is an ordinary expectation against $f_{X|Y}(\cdot \mid y)$. The special cases $g(x, y) = x$ and $g(x, y) = (x - \mathbb{E}[X \mid Y = y])^2$ give the **conditional mean** and **conditional variance**

$$\begin{aligned} \mathbb{E}[X \mid Y = y] &= \int x f_{X|Y}(x \mid y) dx, \\ \text{Var}(X \mid Y = y) &= \mathbb{E}[X^2 \mid Y = y] - (\mathbb{E}[X \mid Y = y])^2. \end{aligned} \quad (73)$$

Both sides of (73) are ordinary functions of the number y . Naming the first of them,

$$\varphi(y) = \mathbb{E}[X \mid Y = y] = \int x f_{X|Y}(x \mid y) dx, \quad (74)$$

$\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is deterministic: fix y and it returns a number. The object written $\mathbb{E}[X \mid Y]$ is that same function evaluated at the *random* argument,

$$\mathbb{E}[X \mid Y] = \varphi(Y), \quad (75)$$

so it is a random variable, and a function of Y in isolation: $\mathbb{E}[X \mid Y = y]$ is a number, while $\mathbb{E}[X \mid Y]$ is a random variable.

Measure-theoretically, $\varphi(Y)$ is $\sigma(Y)$ -measurable, which is the precise sense in which conditioning on Y means using only the information carried by Y .

6.2 Independence

Definition 6.4 (Independence). X and Y are **independent**, written $X \perp Y$, if

$$f(x, y) = f_X(x)f_Y(y) \quad \text{for all } (x, y); \quad (76)$$

equivalently $F(x, y) = F_X(x)F_Y(y)$ for all (x, y) , or $f_{X|Y}(x | y) = f_X(x)$ wherever the conditional is defined.

In the discrete case the last form says that conditioning changes nothing: at any y with $f_Y(y) = \mathbb{P}(Y = y) > 0$, (71) turns $X \perp Y$ into

$$\mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \mathbb{P}(X = x),$$

which is Definition 1.19 applied to the events $\{X = x\}$ and $\{Y = y\}$.

Theorem 6.5 (Factorization criterion). *X and Y are independent if and only if there exist $g(x) \geq 0$ and $h(y) \geq 0$ with $f(x, y) = g(x)h(y)$ for all (x, y) .*

Proof. Necessity is the definition. For sufficiency put $c = \int g$ and $d = \int h$, so $cd = \iint f = 1$; then $f_X(x) = g(x)d$ and $f_Y(y) = h(y)c$, whence $f_X(x)f_Y(y) = cd g(x)h(y) = f(x, y)$. $\square$

The criterion is a statement about the *whole plane*, not just about the support: the hypothesis is $f(x, y) = g(x)h(y)$ for *all* $(x, y) \in \mathbb{R}^2$. Dropping “all” destroys it.

Example 6.6 (Factoring on the support is not enough). Let

$$f(x, y) = 2 \mathbf{1}_{\{0 < x < y < 1\}},$$

a density because the triangle $\{0 < x < y < 1\}$ has area $\frac{1}{2}$. On the triangle f is the constant $2 = 2 \cdot 1$, a product of a function of x and a function of y ; yet X and Y are dependent. The obstruction is the indicator, which couples the two arguments and cannot be factored.

To prove dependence, first the marginals. For $0 < x < 1$ the admissible y are $x < y < 1$, and for $0 < y < 1$ the admissible x are $0 < x < y$, so

$$f_X(x) = \int_x^1 2 dy = 2(1 - x), \quad f_Y(y) = \int_0^y 2 dx = 2y,$$

both densities on $(0, 1)$, as $\int_0^1 2(1 - x)dx = \int_0^1 2y dy = 1$ confirms. Now test (76) at one point. Take $(x, y) = (\frac{3}{4}, \frac{1}{4})$, which lies outside the triangle because $x > y$. There

$$f(x, y) = 0, \quad \text{but} \quad f_X\left(\frac{3}{4}\right) f_Y\left(\frac{1}{4}\right) = 2\left(1 - \frac{3}{4}\right) \cdot 2 \cdot \frac{1}{4} = \frac{1}{4} \neq 0,$$

so (76) fails and $X \not\perp Y$. The same point defeats Theorem 6.5 directly. Suppose $f = gh$ everywhere. Then $g(\frac{3}{4})h(\frac{1}{4}) = f(\frac{3}{4}, \frac{1}{4}) = 0$, so one of the two factors vanishes. But $g(\frac{3}{4}) \neq 0$, because $f(\frac{3}{4}, y) = g(\frac{3}{4})h(y) = 2$ for $y \in (\frac{3}{4}, 1)$; and $h(\frac{1}{4}) \neq 0$, because $f(x, \frac{1}{4}) = g(x)h(\frac{1}{4}) = 2$ for $x \in (0, \frac{1}{4})$. Contradiction.

The conditional distribution makes the dependence visible: for $0 < y < 1$,

$$f_{X|Y}(x | y) = \frac{2 \mathbf{1}\{0 < x < y\}}{2y} = \frac{1}{y} \mathbf{1}\{0 < x < y\},$$

so $X | Y = y$ is uniform on $(0, y)$. The conditional law moves with y , which is exactly what independence forbids: knowing Y tells you an upper bound for X . A support that is a triangle rather than a rectangle is precisely a support on which one coordinate constrains the other.

Theorem 6.7 (Independence and expectations). *If X and Y are independent and the expectations exist, then for all g, h*

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)] \mathbb{E}[h(Y)], \quad (77)$$

and in particular $M_{X+Y}(t) = M_X(t)M_Y(t)$.

Proof. Fubini's theorem applies because the integrand factors and each factor is integrable:

$$\mathbb{E}[g(X)h(Y)] = \iint g(x)h(y)f_X(x)f_Y(y) dx dy = \left(\int g f_X \right) \left(\int h f_Y \right).$$

Take $g(x) = e^{tx}$, $h(y) = e^{ty}$ for the mgf statement. □

Theorem 6.8 (Sums of independent variables). *If $X_1, \dots, X_n$ are independent with mgfs M_i , then $M_{\sum_i X_i}(t) = \prod_{i=1}^n M_i(t)$. Combined with Theorem 3.10(b) this identifies the distribution of the sum.*

Proof. By induction on n . For $n = 2$, independence of X_1 and X_2 makes e^{tX_1} and e^{tX_2} functions of independent variables, so Theorem 6.7 applies with $g(x) = h(x) = e^{tx}$:

$$M_{X_1+X_2}(t) = \mathbb{E}[e^{t(X_1+X_2)}] = \mathbb{E}[e^{tX_1}e^{tX_2}] = \mathbb{E}[e^{tX_1}] \mathbb{E}[e^{tX_2}] = M_1(t)M_2(t),$$

valid for every t in the intersection of the two neighbourhoods of 0 on which the mgfs exist, itself a neighbourhood of 0.

Suppose the claim holds for n . Write $S_n = \sum_{i=1}^n X_i$. Then S_n is a function of $X_1, \dots, X_n$ alone and X_{n+1} is independent of that collection, so S_n and X_{n+1} are independent and the case $n = 2$ gives

$$M_{S_{n+1}}(t) = M_{S_n}(t) M_{X_{n+1}}(t) = \left(\prod_{i=1}^n M_i(t) \right) M_{X_{n+1}}(t) = \prod_{i=1}^{n+1} M_i(t),$$

the middle equality by the induction hypothesis. $\square$

Applying it to Table 2 and Table 3: sums of independent Bernoulli(p) are binomial; of independent Poisson(λ_i) are Poisson($\sum_i \lambda_i$); of independent Gamma(α_i, β) are Gamma($\sum_i \alpha_i, \beta$); and of independent $N(\mu_i, \sigma_i^2)$ are $N(\sum_i \mu_i, \sum_i \sigma_i^2)$.

6.3 Hierarchical models and iterated expectations

Theorem 6.9 (Law of iterated expectations). *If $\mathbb{E}[|X|] < \infty$ then*

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]]. \quad (78)$$

Proof. Write the outer expectation over Y and unfold the inner one, then exchange the order of integration (Fubini, legitimate since $\mathbb{E}[|X|] < \infty$):

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X | Y]] &= \int \left(\int x f(x | y) dx \right) f_Y(y) dy \\ &= \iint x f(x, y) dx dy = \int x f_X(x) dx = \mathbb{E}[X], \end{aligned}$$

using $f(x | y)f_Y(y) = f(x, y)$ and (68). $\square$

Theorem 6.10 (Conditional variance identity).

$$\text{Var } X = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]). \quad (79)$$

Proof. Write the two conditional moments as the random variables they are, in the notation of (75):

$$\varphi(Y) = \mathbb{E}[X | Y], \quad \psi(Y) = \text{Var}(X | Y).$$

Step 1: the conditional second moment. Applying (41) to the conditional distribution of X given $Y = y$, which is a bona fide distribution by Definition 6.1, gives $\text{Var}(X | Y = y) = \mathbb{E}[X^2 | Y = y] - (\mathbb{E}[X | Y = y])^2$, that is

$$\mathbb{E}[X^2 | Y] = \psi(Y) + \varphi(Y)^2. \quad (80)$$

Step 2: unconditional moments. Apply (78) twice, once to X^2 and once to X , and use (80) together with linearity (Theorem 3.2(a)):

$$\begin{aligned} \mathbb{E}[X^2] &= \mathbb{E}[\mathbb{E}[X^2 | Y]] = \mathbb{E}[\psi(Y) + \varphi(Y)^2] = \mathbb{E}[\psi(Y)] + \mathbb{E}[\varphi(Y)^2], \\ \mathbb{E}[X] &= \mathbb{E}[\varphi(Y)]. \end{aligned}$$

Step 3: subtract. By (41) applied to X ,

$$\text{Var } X = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathbb{E}[\psi(Y)] + \underbrace{\mathbb{E}[\varphi(Y)^2] - (\mathbb{E}[\varphi(Y)])^2}_{= \text{Var } \varphi(Y)},$$

the bracket being (41) once more, this time applied to the random variable $\varphi(Y)$. Since $\mathbb{E}[\psi(Y)] = \mathbb{E}[\text{Var}(X | Y)]$ and $\text{Var } \varphi(Y) = \text{Var}(\mathbb{E}[X | Y])$, this is (79). $\square$

Read (79) as a decomposition of variation into a part left unexplained by Y , namely $\mathbb{E}[\text{Var}(X | Y)]$, and a part explained by Y , namely $\text{Var}(\mathbb{E}[X | Y])$. Both terms are nonnegative, so each is a lower bound on $\text{Var } X$: conditioning can only reduce variance on average.

Example 6.11 (A hierarchical model). Take a Poisson count whose rate is itself random: $X | \Lambda \sim \text{Poisson}(\Lambda)$ with $\Lambda \sim \text{Gamma}(\alpha, \beta)$. By Table 2 and Table 3,

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[\mathbb{E}[X | \Lambda]] = \mathbb{E}[\Lambda] = \alpha\beta, \\ \text{Var } X &= \mathbb{E}[\text{Var}(X | \Lambda)] + \text{Var}(\mathbb{E}[X | \Lambda]) = \mathbb{E}[\Lambda] + \text{Var } \Lambda = \alpha\beta(1 + \beta) > \mathbb{E}[X]. \end{aligned}$$

Example 6.12 (Iterated expectations needs integrability). Let Z_1 and Z_2 be independent $N(0, 1)$ variables and put

$$X = \frac{Z_1}{Z_2} \mathbf{1}\{Z_2 \neq 0\}.$$

The indicator makes X defined at every sample point, and since $\mathbb{P}(Z_2 = 0) = 0$ by Proposition 2.17 it does not affect the distribution of X .

Condition on Z_2 . On $\{Z_2 = 0\}$ we have $X = 0$, so $\mathbb{E}[X \mid Z_2 = 0] = 0$. For $y \neq 0$, fixing $Z_2 = y$ leaves $X = Z_1/y$, and $Z_1 \perp Z_2$ means the conditional distribution of Z_1 given $Z_2 = y$ is its unconditional $N(0, 1)$ (Definition 6.4), so

$$\mathbb{E}[X \mid Z_2 = y] = \frac{1}{y} \mathbb{E}[Z_1] = 0.$$

Hence $\varphi(y) = 0$ for every y , so $\mathbb{E}[X \mid Z_2] = 0$ identically and

$$\mathbb{E}[\mathbb{E}[X \mid Z_2]] = \mathbb{E}[0] = 0. \quad (81)$$

But X has no mean. Apply Theorem B.4 to $(U, V) = (Z_1/Z_2, Z_2)$, a bijection off the null set $\{z_2 = 0\}$, with inverse $z_1 = uv$, $z_2 = v$ and

$$J = \det \begin{pmatrix} v & u \\ 0 & 1 \end{pmatrix} = v.$$

Since $Z_1 \perp Z_2$ their joint density is the product $(2\pi)^{-1} e^{-(z_1^2 + z_2^2)/2}$, so

$$f_{U,V}(u, v) = \frac{1}{2\pi} e^{-(u^2 v^2 + v^2)/2} |v|,$$

and integrating v out, the integrand being even in v ,

$$\begin{aligned} f_X(u) &= \int_{-\infty}^{\infty} f_{U,V}(u, v) dv = \frac{1}{\pi} \int_0^{\infty} v e^{-v^2(1+u^2)/2} dv \\ &= \frac{1}{\pi} \left[\frac{-e^{-v^2(1+u^2)/2}}{1+u^2} \right]_0^{\infty} = \frac{1}{\pi} \frac{1}{1+u^2}, \end{aligned}$$

the standard Cauchy density of Table 3. By (38), $\mathbb{E}[|X|] = \infty$, so $\mathbb{E}[X]$ does not exist.

So the right-hand side of (78) is perfectly well defined and equal to 0, while the left-hand side does not exist: the hypothesis $\mathbb{E}[|X|] < \infty$ in Theorem 6.9 is not decorative.

7 Lecture 7: Random Samples

7.1 Random samples and statistics

Definition 7.1 (Random sample). $X_1, \dots, X_n$ are a **random sample** of size n from the population f if they are mutually independent and each has marginal pmf or pdf f . Equivalently they are **independent and identically distributed**, written $X_i \sim \text{iid } f$. The joint pmf or pdf is then

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i), \quad \text{or} \quad f(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^n f(x_i \mid \theta) \quad (82)$$

when the population belongs to a parametric family, the same θ appearing in every factor.

Draws *without* replacement from a finite population are identically distributed but dependent; the iid model is then an approximation, good when n is small relative to the population size [Casella and Berger, 2024].

Remark 7.2 (Continuous coordinates, no joint density). Independence is what delivers the product (82). Without it, a vector whose coordinates are each continuous need not be continuous. Let $X \sim N(0, 1)$, with cdf Φ and pdf φ as in (43).

- (i) $(X, X)^\top$ lies in the diagonal $D = \{(x, x) : x \in \mathbb{R}\}$, which has zero area. If it had a joint pdf f , (64) would give $1 = \mathbb{P}((X, X)^\top \in D) = \int_D f = 0$.
- (ii) Let $Z \sim \text{Bernoulli}(\frac{1}{2})$ be independent of X , and $Y = X + Z$. Partitioning on $\{Z = 0\}$ and $\{Z = 1\}$ (Theorem 1.12(a)) and using independence,

$$\begin{aligned} F_Y(y) &= \frac{1}{2}\mathbb{P}(X \leq y) + \frac{1}{2}\mathbb{P}(X \leq y - 1) = \frac{1}{2}\Phi(y) + \frac{1}{2}\Phi(y - 1) \\ &= \int_{-\infty}^y \frac{1}{2}[\varphi(s) + \varphi(s - 1)] ds, \end{aligned}$$

so Y is continuous. Yet $(X, Y)^\top$ lies with probability one in $L = \{(x, y) : y - x \in \{0, 1\}\}$, a union of two lines with zero area, so the argument in (i) shows that $(X, Y)^\top$ has no joint pdf.

Definition 7.3 (Statistic; sampling distribution). Let $X_1, \dots, X_n$ be a random sample and $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ a function that does not involve any unknown parameter. The

random vector $T(X_1, \dots, X_n)$ is a **statistic**, and its distribution is its **sampling distribution**. The leading examples are the **sample mean** and the **sample variance**,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad (83)$$

and the **sample standard deviation** $S = \sqrt{S^2}$. Lowercase $\bar{x}$, s^2 , s denote observed values.

Theorem 7.4 (Sum-of-squares identity). *For any numbers $x_1, \dots, x_n$ and any $a \in \mathbb{R}$,*

$$\sum_{i=1}^n (x_i - a)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - a)^2. \quad (84)$$

Hence:

$$(a) \min_{a \in \mathbb{R}} \sum_i (x_i - a)^2 = \sum_i (x_i - \bar{x})^2, \text{ attained only at } a = \bar{x};$$

$$(b) (n-1)s^2 = \sum_i x_i^2 - n\bar{x}^2.$$

Proof. Write $x_i - a = (x_i - \bar{x}) + (\bar{x} - a)$ and expand. The cross term $2(\bar{x} - a) \sum_i (x_i - \bar{x})$ vanishes because $\sum_i (x_i - \bar{x}) = 0$. Part (a) is read off, and (b) is $a = 0$. $\square$

7.2 Moments of the sample mean and variance

Recall from Section B.1 that covariance is symmetric and bilinear, and that

$$\text{Var } X = \text{Cov}(X, X), \quad \text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

and $\text{Cov}(X, Y) = 0$ if $X \perp Y$.

Proposition 7.5 (Sums over a random sample). *If $X_1, \dots, X_n$ is a random sample and $\mathbb{E}[g(X_1)]$ and $\text{Var } g(X_1)$ exist, then*

$$\mathbb{E} \left[\sum_{i=1}^n g(X_i) \right] = n \mathbb{E}[g(X_1)], \quad \text{Var} \left(\sum_{i=1}^n g(X_i) \right) = n \text{Var } g(X_1).$$

Proof. The mean is linearity plus identical distribution. For the variance, the bilinearity of covariance extends Theorem B.2 to

$$\text{Var}\left(\sum_{i=1}^n Y_i\right) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(Y_i, Y_j) = \sum_{i=1}^n \text{Var } Y_i + \sum_{i \neq j} \text{Cov}(Y_i, Y_j). \quad (85)$$

With $Y_i = g(X_i)$ the Y_i are independent, so each covariance with $i \neq j$ is zero (Theorem B.2), and the n variances are equal. $\square$

Theorem 7.6 (Mean and variance of $\bar{X}$ and S^2). *Let $X_1, \dots, X_n$ be a random sample from a population with mean μ and variance $\sigma^2 < \infty$, written $X_i \sim \text{iid}(\mu, \sigma^2)$. Then*

$$(a) \mathbb{E}[\bar{X}] = \mu, \quad (b) \text{Var } \bar{X} = \frac{\sigma^2}{n}, \quad (c) \mathbb{E}[S^2] = \sigma^2 \text{ if } n \geq 2.$$

Proof. (a) By linearity (Theorem 3.2(a)),

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \cdot n\mu = \mu.$$

(b) By bilinearity, splitting the double sum into the terms $j = i$ and $j \neq i$,

$$\begin{aligned} \text{Var } \bar{X} &= \text{Cov}\left(\frac{1}{n} \sum_{i=1}^n X_i, \frac{1}{n} \sum_{j=1}^n X_j\right) = \sum_{i=1}^n \left\{ \text{Cov}\left(\frac{X_i}{n}, \frac{X_i}{n}\right) + \sum_{j \neq i} \text{Cov}\left(\frac{X_i}{n}, \frac{X_j}{n}\right) \right\} \\ &= \sum_{i=1}^n \text{Var}\left(\frac{X_i}{n}\right) = n \cdot \frac{\sigma^2}{n^2} = \frac{\sigma^2}{n}, \end{aligned}$$

since the terms with $j \neq i$ vanish by independence and $\text{Var}(X_i/n) = \sigma^2/n^2$ by (42).

(c) By Theorem 7.4(b), $(n-1)S^2 = \sum_i X_i^2 - n\bar{X}^2$. By (41), (a) and (b),

$$\mathbb{E}[X_i^2] = \text{Var } X_i + (\mathbb{E}[X_i])^2 = \sigma^2 + \mu^2, \quad \mathbb{E}[\bar{X}^2] = \text{Var } \bar{X} + (\mathbb{E}[\bar{X}])^2 = \frac{\sigma^2}{n} + \mu^2.$$

Taking expectations, dividing by $n-1 \geq 1$ and using linearity,

$$\mathbb{E}[S^2] = \frac{1}{n-1} \left(\sum_{i=1}^n \mathbb{E}[X_i^2] - n\mathbb{E}[\bar{X}^2] \right) = \frac{n(\sigma^2 + \mu^2) - n(\sigma^2/n + \mu^2)}{n-1} = \sigma^2. \quad \square$$

Why the divisor is $n - 1$. Let $X_i \sim \text{iid}(\mu, \sigma^2)$ with $n \geq 2$.

Step 1. By (84) with $a = \mu$,

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2.$$

Step 2. The two outer terms have known means: $\mathbb{E}[(X_i - \mu)^2] = \text{Var } X_i = \sigma^2$, so $\mathbb{E}[\sum_i (X_i - \mu)^2] = n\sigma^2$; and, by (a) and (b), $\mathbb{E}[n(\bar{X} - \mu)^2] = n \text{Var } \bar{X} = \sigma^2$.

Step 3. Taking expectations in Step 1 and using linearity, $n\sigma^2 = \mathbb{E}[\sum_i (X_i - \bar{X})^2] + \sigma^2$, that is,

$$\mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = (n - 1)\sigma^2.$$

Step 4. Dividing by $n - 1$ gives $\mathbb{E}[S^2] = \sigma^2$, while dividing by n gives

$$\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right] = \frac{n - 1}{n} \sigma^2 < \sigma^2.$$

Deviations are measured from $\bar{X}$, the minimizer in Theorem 7.4(a), rather than from μ , and on average this costs exactly one σ^2 .

7.3 The sampling distribution of the sample mean

Theorem 7.7 (mgf of the sample mean). *If $X_1, \dots, X_n$ is a random sample from a population with mgf M_X , then $M_{\bar{X}}(t) = [M_X(t/n)]^n$.*

Proof. Theorem 3.12 with $a = 1/n$ gives $M_{\bar{X}}(t) = M_{\sum_i X_i}(t/n)$, and Theorem 6.8 with n identical factors gives $[M_X(t/n)]^n$. $\square$

Example 7.8 (Normal and gamma samples). For $X_i \sim \text{iid } N(\mu, \sigma^2)$, Proposition 4.7 gives

$$M_{\bar{X}}(t) = \left[\exp\left(\frac{\mu t}{n} + \frac{\sigma^2 t^2}{2n^2}\right) \right]^n = \exp\left(\mu t + \frac{(\sigma^2/n) t^2}{2}\right),$$

so $\bar{X} \sim N(\mu, \sigma^2/n)$ by Theorem 3.10(b). The same computation with the $\Gamma(\alpha, \beta)$ mgf $(1 - \beta t)^{-\alpha}$, $t < 1/\beta$, gives $\bar{X} \sim \Gamma(n\alpha, \beta/n)$.⁴

⁴Written after $\sim$, $\Gamma(\alpha, \beta)$ and $B(\alpha, \beta)$ denote the gamma and beta distributions of Table 3.

7.4 Sampling from the normal distribution

Proposition 7.9 (Normal and chi squared facts). *Write χ_p^2 for the chi squared distribution with p degrees of freedom, that is, $\Gamma(p/2, 2)$ (Table 3).*

- (a) *If $X_1, \dots, X_m$ are independent with $X_i \sim N(\mu_i, \sigma_i^2)$, and $a_1, \dots, a_m \in \mathbb{R}$ are not all zero, then $\sum_i a_i X_i \sim N(\sum_i a_i \mu_i, \sum_i a_i^2 \sigma_i^2)$.*
- (b) *If $Z \sim N(0, 1)$ then $Z^2 \sim \chi_1^2$.*
- (c) *If $V_1, \dots, V_m$ are independent with $V_i \sim \chi_{p_i}^2$, then $\sum_i V_i \sim \chi_{p_1 + \dots + p_m}^2$.*

Proof. (a) By Theorem 3.12 and Proposition 4.7, $M_{a_i X_i}(t) = M_{X_i}(a_i t) = \exp(a_i \mu_i t + a_i^2 \sigma_i^2 t^2 / 2)$. The $a_i X_i$ are independent, so Theorem 6.8 gives

$$M_{\sum_i a_i X_i}(t) = \exp\left(\left(\sum_i a_i \mu_i\right)t + \left(\sum_i a_i^2 \sigma_i^2\right)\frac{t^2}{2}\right),$$

the mgf of the stated normal; conclude by Theorem 3.10(b). (b) is Example 2.24. (c) The χ_p^2 mgf is $(1 - 2t)^{-p/2}$ for $t < \frac{1}{2}$; multiply the mgfs (Theorem 6.8) and apply Theorem 3.10(b). $\square$

Theorem 7.10 (Sampling from the normal; proof omitted). *Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$, with $n \geq 2$. Then*

- (a) *$\bar{X}$ and S^2 are independent;*
- (b) *$\bar{X} \sim N(\mu, \sigma^2/n)$;*
- (c) *$(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$.*

See Casella and Berger [2024, Thm. 5.3.1]: (b) is Proposition 7.9(a), (a) follows from Theorem B.10(a) and (c) applied to $(\bar{X}, X_2 - \bar{X}, \dots, X_n - \bar{X})$, and (c) from (84) with $a = \mu$ and an mgf argument. By (b), $(\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$, which is of no direct use when σ is unknown. Replacing σ by S leads to Student's t .

Definition 7.11 (Student's t). If $Z \sim N(0, 1)$ and $V \sim \chi_p^2$ are independent, then $T = Z/\sqrt{V/p}$ has **Student's t distribution with p degrees of freedom**, written $T \sim t_p$, with pdf⁵

$$f_T(t) = \frac{\Gamma(\frac{p+1}{2})}{\Gamma(\frac{p}{2})} \frac{1}{\sqrt{p\pi}} \frac{1}{(1 + t^2/p)^{(p+1)/2}}, \quad t \in \mathbb{R}. \quad (86)$$

Corollary 7.12 (The t statistic). Under Theorem 7.10,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Proof. Divide numerator and denominator by $\sigma/\sqrt{n}$:

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{[(n-1)S^2/\sigma^2]/(n-1)}}.$$

The numerator is $N(0, 1)$ by Theorem 7.10(b), the bracket is χ_{n-1}^2 by (c), and the two are independent by (a). Apply Definition 7.11. $\square$

Definition 7.13 (Snedecor's F). If $U \sim \chi_p^2$ and $V \sim \chi_q^2$ are independent, then $F = (U/p)/(V/q)$ has the **F distribution with p and q degrees of freedom**, written $F \sim F_{p,q}$, with pdf

$$f_F(x) = \frac{\Gamma(\frac{p+q}{2})}{\Gamma(\frac{p}{2}) \Gamma(\frac{q}{2})} \left(\frac{p}{q}\right)^{p/2} \frac{x^{p/2-1}}{[1 + (p/q)x]^{(p+q)/2}}, \quad x > 0. \quad (87)$$

Its use: for independent samples of sizes n and m from $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$, Theorem 7.10(c) applied to each sample gives

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{n-1, m-1},$$

a distribution free of all four parameters, used for inference on σ_X^2/σ_Y^2 . Also $\mathbb{E}[F] = q/(q-2)$ for $q > 2$.

⁵The density follows from Theorem B.4 applied to $(t, w) = (z/\sqrt{v/p}, v)$ and integrating out w against a gamma kernel [Casella and Berger, 2024, Def. 5.3.4]. Facts [Casella and Berger, 2024, Exercise 5.18]: t_1 is the Cauchy; t_p has moments only of order less than p , hence no mgf; $\mathbb{E}[T] = 0$ for $p > 1$ and $\text{Var } T = p/(p-2)$ for $p > 2$; and $f_T \rightarrow$ the $N(0, 1)$ density as $p \rightarrow \infty$. The tails are heavier than the normal's.

Theorem 7.14 (Relations among t , F , and beta).

- (a) If $X \sim F_{p,q}$ then $1/X \sim F_{q,p}$.
- (b) If $X \sim t_q$ then $X^2 \sim F_{1,q}$.
- (c) If $X \sim F_{p,q}$ then $\frac{(p/q)X}{1 + (p/q)X} \sim B(p/2, q/2)$.

Proof omitted [Casella and Berger, 2024, Exercises 5.17–5.18].

7.5 Order statistics

Definition 7.15 (Order statistics). The **order statistics** of a random sample are its values placed in ascending order,

$$X_{(1)} = \min_{1 \leq i \leq n} X_i \leq X_{(2)} \leq \cdots \leq X_{(n)} = \max_{1 \leq i \leq n} X_i.$$

Statistics built from them include the **sample range** $R = X_{(n)} - X_{(1)}$, the **sample median**

$$M = \begin{cases} X_{((n+1)/2)}, & n \text{ odd}, \\ \frac{1}{2} (X_{(n/2)} + X_{(n/2+1)}), & n \text{ even}, \end{cases} \quad (88)$$

and, for $p \in (0, 1)$, the $(100p)$ th **sample percentile**

$$X_{(\{np\})} \text{ if } \frac{1}{2n} < p < \frac{1}{2}, \quad X_{(n+1-\{n(1-p)\})} \text{ if } \frac{1}{2} < p < 1 - \frac{1}{2n},$$

where $\{b\}$ is b rounded to the nearest integer: $\{b\} = i$ for the integer i with $i - \frac{1}{2} \leq b < i + \frac{1}{2}$.

Theorem 7.16 (cdf of an order statistic). Let $X_i \sim \text{iid } F$, where F is any cdf. For $j = 1, \dots, n$ and $x \in \mathbb{R}$,

$$F_{X_{(j)}}(x) = \sum_{k=j}^n \binom{n}{k} F(x)^k [1 - F(x)]^{n-k}. \quad (89)$$

In particular $F_{X_{(n)}}(x) = F(x)^n$ and $F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$.

Proof. Fix x and let $Y = \sum_i \mathbf{1}\{X_i \leq x\}$ count the observations at or below x . The indicators are iid Bernoulli($F(x)$), so $Y \sim \text{binomial}(n, F(x))$. Since $X_{(j)} \leq x$ exactly

when at least j observations are at most x , $\{X_{(j)} \leq x\} = \{Y \geq j\}$, which is (89). For the maximum this reads

$$\mathbb{P}(X_{(n)} \leq x) = \mathbb{P}(X_1 \leq x, \dots, X_n \leq x) = \prod_{i=1}^n \mathbb{P}(X_i \leq x) = F(x)^n,$$

and for the minimum, $\mathbb{P}(X_{(1)} > x) = \mathbb{P}(X_1 > x, \dots, X_n > x) = [1 - F(x)]^n$. $\square$

For a discrete population with support $x_1 < x_2 < \dots$ the pmf is the jump $\mathbb{P}(X_{(j)} = x_i) = F_{X_{(j)}}(x_i) - F_{X_{(j)}}(x_{i-1})$ [Casella and Berger, 2024, Thm. 5.4.3].

Theorem 7.17 (Densities of order statistics). *Let $X_i \sim iid$ with continuous cdf F and pdf f .*

(a) For $j = 1, \dots, n$,

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} f(x) F(x)^{j-1} [1 - F(x)]^{n-j}. \quad (90)$$

(b) For $1 \leq i < j \leq n$ and $u < v$,

$$f_{X_{(i)}, X_{(j)}}(u, v) = \frac{n!}{(i-1)!(j-1-i)!(n-j)!} f(u)f(v) F(u)^{i-1} \times [F(v) - F(u)]^{j-1-i} [1 - F(v)]^{n-j},$$

and 0 for $u \geq v$.

(c) $f_{X_{(1)}, \dots, X_{(n)}}(x_1, \dots, x_n) = n! \prod_{i=1}^n f(x_i)$ for $x_1 < \dots < x_n$, and 0 otherwise.

8 Lecture 8: Point Estimation

8.1 Large-sample approximations: a preview

Theorem 7.10 is exact but requires normality. Without it, for $X_i \sim \text{iid}(\mu, \sigma^2)$ and n large,

$$\bar{X} \stackrel{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \quad [\text{CLT}], \quad S^2 \approx \sigma^2 \quad [\text{LLN}].$$

A sequence Y_n converges **in probability** to Y , written $Y_n \rightarrow_p Y$, if $\mathbb{P}(|Y_n - Y| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$. It converges **in distribution**, written $Y_n \rightarrow_d Y$, if $F_{Y_n}(y) \rightarrow F_Y(y)$ at every continuity point y of F_Y .

Theorem 8.1 (Weak law of large numbers). *If $X_i \sim \text{iid}(\mu, \sigma^2)$ with $\sigma^2 < \infty$, then $\bar{X}_n \rightarrow_p \mu$.*

Proof. By Chebyshev's inequality⁶ and Theorem 7.6(b),

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

□

Theorem 8.2 (Central limit theorem). *If $X_i \sim \text{iid}(\mu, \sigma^2)$ with $0 < \sigma^2 < \infty$, then*

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \rightarrow_d N(0, 1).$$

Proof. Assume the mgf of X_1 is finite near 0; otherwise the same argument runs with characteristic functions [Casella and Berger, 2024, Thm. 5.5.15]. The $Y_i = (X_i - \mu)/\sigma$ are iid with $\mathbb{E}[Y_1] = 0$, $\mathbb{E}[Y_1^2] = 1$ and mgf $M(s)$ finite for $|s| < h$, and $\sqrt{n}(\bar{X}_n - \mu)/\sigma = n^{-1/2} \sum_i Y_i$ has mgf $M_n(t) = [M(t/\sqrt{n})]^n$, $|t| < h\sqrt{n}$, by Theorems 3.12 and 6.7. As $M(0) = 1$, $M'(0) = 0$ and $M''(0) = 1$, Taylor's theorem gives $M(s) = 1 + s^2/2 + o(s^2)$, so with $\log(1 + x) = x + O(x^2)$,

$$\log M_n(t) = n \log \left(1 + \frac{t^2}{2n} + o(n^{-1}) \right) \rightarrow \frac{t^2}{2}$$

for fixed t . Hence $M_n(t) \rightarrow e^{t^2/2}$, the $N(0, 1)$ mgf (56), and Theorem 3.11 concludes.

□

⁶If $\text{Var } Y < \infty$ and $\varepsilon > 0$, then $\mathbb{P}(|Y - \mathbb{E}[Y]| \geq \varepsilon) \leq \text{Var } Y / \varepsilon^2$.

8.2 From probability to statistics

Statistics runs probability backwards:

- **data:** a vector $\mathbf{x} = (x_1, \dots, x_n)^\top$ of observations;
- **assumption:** $\mathbf{x}$ is a realization of a random vector $\mathbf{X} = (X_1, \dots, X_n)^\top$, that is, $\mathbf{X}(s) = \mathbf{x}$ for some outcome $s \in S$;
- **objective:** use $\mathbf{x}$ to give data-based answers to questions about the distribution of $\mathbf{X}$.

In probability the distribution is known and the outcome is not; in statistics the outcome is known and the distribution is not.

The setup in this course is *parametric*: $X_1, \dots, X_n$ is a random sample from a pmf or pdf $f(\cdot \mid \theta)$ of known form, with $\theta \in \Theta$ unknown. For $\theta \in \Theta$, $\mathbb{P}_\theta$, $\mathbb{E}_\theta$ and Var_θ denote probability, expectation and variance under the joint pmf or pdf $\prod_i f(x_i \mid \theta)$ of (82). Three running examples are

- (1) $X_i \sim \text{iid Bernoulli}(p)$, with $p \in [0, 1]$ unknown;
- (2) $X_i \sim \text{iid U}[0, \theta]$, with $\theta > 0$ unknown;
- (3) $X_i \sim \text{iid N}(\mu, \sigma^2)$, with $\mu \in \mathbb{R}$ and/or $\sigma^2 > 0$ unknown.

Table 5: How tractable the three models are, from the probability side and from the statistics side.

	Probability tractable	Probability intractable
Statistics tractable	(1) Bernoulli	(3) Normal
Statistics intractable	(2) U[0, θ]	?

Bernoulli is tractable on both sides. The uniform has simple probabilities but support that depends on θ ; the normal has no closed-form cdf but exact sampling distributions. The three types of inference are

- (1) **point estimation** [Casella and Berger, 2024, Ch. 7]: what is θ ?
- (2) **hypothesis testing** [Casella and Berger, 2024, Ch. 8]: is $\theta = \theta_0$?
- (3) **interval estimation** [Casella and Berger, 2024, Ch. 9]: what are the plausible values of θ ?

8.3 Point estimators and the method of moments

Definition 8.3 (Point estimator). A **point estimator** of θ is any statistic $\hat{\theta} = W(X_1, \dots, X_n)$, with W a function that does not depend on θ . Its value at the realized outcome $s \in S$, $W(X_1, \dots, X_n)(s) = W(X_1(s), \dots, X_n(s)) = W(x_1, \dots, x_n)$, is an **estimate**.

Nothing in the definition requires $\hat{\theta}$ to take values in Θ . A good estimator is one that is close to θ in some probabilistic sense.

Idea of the method of moments. The population moments $\mathbb{E}_\theta[X_1], \mathbb{E}_\theta[X_1^2], \dots$ are known functions of the unknown θ , while their sample counterparts can be computed from the data and, by the law of large numbers (Theorem 8.1), are close to them when n is large. Equating the two and solving for θ gives an estimator.

Definition 8.4 (Method of moments).

- (a) *Scalar case.* Let $\Theta \subseteq \mathbb{R}$, and let $\mu(\theta) = \mathbb{E}_\theta[X_1]$ be the population mean, assumed to exist for every $\theta \in \Theta$. A **method of moments estimator** $\hat{\theta}_{\text{MM}}$ is a solution $\hat{\theta}_{\text{MM}} \in \Theta$ of

$$\mu(\hat{\theta}_{\text{MM}}) = \bar{X}. \quad (91)$$

- (b) *General case.* Let $\Theta \subseteq \mathbb{R}^k$, so that θ has k unknown coordinates. For $j = 1, \dots, k$:

- X_i^j is the j th power of X_i , where $x^j = \prod_{l=1}^j x$ for $x \in \mathbb{R}$;
- the **j th population moment** is the function $\mu_j : \Theta \rightarrow \mathbb{R}$ given by

$$\mu_j(\theta) = \mathbb{E}_\theta[X_1^j] = \begin{cases} \sum_x x^j f(x | \theta), & X_i \text{ discrete,} \\ \int_{-\infty}^{\infty} x^j f(x | \theta) dx, & X_i \text{ continuous,} \end{cases}$$

assumed to exist for every $\theta \in \Theta$ (it is the same for every i , since the X_i are identically distributed);

- the **j th sample moment** is the statistic $m_j = \frac{1}{n} \sum_{i=1}^n X_i^j$.

A **method of moments estimator** $\hat{\theta}_{\text{MM}}$ is a solution $\hat{\theta}_{\text{MM}} \in \Theta$ of the k equations in k unknowns

$$\mu_j(\hat{\theta}_{\text{MM}}) = m_j, \quad j = 1, \dots, k. \quad (92)$$

So j indexes the moment, and one uses as many moments as there are unknown parameters; (a) is (b) with $k = 1$, since $\mu_1 = \mu$ and $m_1 = \bar{X}$. Existence of the moments, and existence (typically also uniqueness) of a solution of (92), are assumptions; $\hat{\theta}_{\text{MM}}$ depends on the sample only through $m_1, \dots, m_k$.

Example 8.5 (Method of moments in the three models). Each case has two steps: the *moment equations* compute $\mu_j(\theta)$; the *estimation equations* set $\mu_j(\hat{\theta}_{\text{MM}}) = m_j$ and solve.

(1) *Bernoulli*(p), $k = 1$. Moment equation: $\mu(p) = \mathbb{E}_p[X_1] = 0 \cdot (1 - p) + 1 \cdot p = p$. Estimation equation: $\hat{p}_{\text{MM}} = \bar{X}$.

(2) $U[0, \theta]$, $k = 1$. Moment equation: $\mu(\theta) = \mathbb{E}_\theta[X_1] = \int_0^\theta x \theta^{-1} dx = \theta/2$. Estimation equation: $\hat{\theta}_{\text{MM}}/2 = \bar{X}$, so $\hat{\theta}_{\text{MM}} = 2\bar{X}$. The model is *non-regular*: its support $[0, \theta]$ depends on θ , so $X_i > \theta$ is impossible. Yet $\hat{\theta}_{\text{MM}}$ can be smaller than an observation: for the sample $\mathbf{x} = (0.1, 0.2, 0.9)$, $\hat{\theta}_{\text{MM}} = 2(0.4) = 0.8 < 0.9 = x_{(3)}$, an estimate that the data themselves rule out. (For $n \leq 2$ this cannot happen, since then $2\bar{X} \geq X_{(n)}$.) Hence $\max(\hat{\theta}_{\text{MM}}, X_{(n)})$ “should” be better; Example 10.8 finds the best unbiased estimator.

(3) $N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$, $k = 2$. Moment equations: $\mu_1(\theta) = \mathbb{E}_\theta[X_1] = \mu$ and $\mu_2(\theta) = \mathbb{E}_\theta[X_1^2] = \text{Var}_\theta(X_1) + (\mathbb{E}_\theta[X_1])^2 = \sigma^2 + \mu^2$. Estimation equations: $\hat{\mu}_{\text{MM}} = m_1$ and $\hat{\sigma}_{\text{MM}}^2 + \hat{\mu}_{\text{MM}}^2 = m_2$. Solving, and using Theorem 7.4(b),

$$\hat{\mu}_{\text{MM}} = \bar{X}, \quad \hat{\sigma}_{\text{MM}}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2.$$

9 Lecture 9: Maximum Likelihood and Mean Squared Error

9.1 Maximum likelihood

Definition 9.1 (Likelihood; maximum likelihood estimator). For $\theta \in \Theta$, the **likelihood function** and the **log likelihood function** are

$$L(\theta \mid X_1, \dots, X_n) = \prod_{i=1}^n f(X_i \mid \theta), \quad \ell(\theta \mid X_1, \dots, X_n) = \sum_{i=1}^n \log f(X_i \mid \theta). \quad (93)$$

For each θ , $L(\theta \mid X_1, \dots, X_n) : S \rightarrow [0, \infty)$ is a random variable, evaluated at the realized outcome $s \in S$:

$$L(\theta \mid X_1, \dots, X_n)(s) = L(\theta \mid \mathbf{x}) = \prod_{i=1}^n f(x_i \mid \theta), \quad x_i = X_i(s),$$

the joint pmf or pdf (82) of the observed data read as a function of θ ; similarly for ℓ . A **maximum likelihood estimator** (MLE) is a statistic $\hat{\theta}_{\text{ML}} = \hat{\theta}(X_1, \dots, X_n)$ such that, for every $\mathbf{x}$, $\hat{\theta}(\mathbf{x}) \in \Theta$ and $L(\hat{\theta}(\mathbf{x}) \mid \mathbf{x}) = \sup_{\theta \in \Theta} L(\theta \mid \mathbf{x})$: at each outcome s , $\hat{\theta}_{\text{ML}}(s)$ maximizes the realized likelihood.

Since log is strictly increasing, L and ℓ have the same maximizers. It is enough that the maximizer exist for all $\mathbf{x}$ in a set of $\mathbb{P}_\theta$ -probability one for every θ . Under regularity conditions $\hat{\theta}_{\text{ML}}$ is optimal in large samples [Casella and Berger, 2024, §10.1].

Example 9.2 (Maximum likelihood: Bernoulli). Let $X_i \sim \text{iid Bernoulli}(p)$, $\Theta = [0, 1]$, and write $T = \sum_{i=1}^n X_i$.

Step 1. *The density.* For $x \in \{0, 1\}$ and $0 < p < 1$, in the exponential-family form (59) of Example 5.4,

$$f(x \mid p) = p^x(1-p)^{1-x} = \underbrace{(1-p)}_{c(p)} \exp\left(\underbrace{\log \frac{p}{1-p}}_{w(p)} \underbrace{x}_{t(x)}\right).$$

Step 2. *The likelihood.* Multiplying over the sample,

$$\begin{aligned} L(p \mid X_1, \dots, X_n) &= \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i} = p^T (1-p)^{n-T} \\ &= (1-p)^n \exp\left(T \log \frac{p}{1-p}\right), \end{aligned}$$

with the convention $0^0 = 1$: the data enter only through the sufficient statistic T (Theorem 10.3), and at the outcome s the likelihood is evaluated at $x_i = X_i(s)$.

Step 3. *The log likelihood.* For $0 < p < 1$,

$$\ell(p \mid X_1, \dots, X_n) = T \log p + (n - T) \log(1 - p) = n \log(1 - p) + T \log \frac{p}{1-p}.$$

Step 4. *The maximum.* Assume $0 < T < n$, so that $\ell \rightarrow -\infty$ at $p = 0$ and at $p = 1$. On $(0, 1)$,

$$\frac{\partial \ell}{\partial p} = \frac{T}{p} - \frac{n - T}{1 - p} = \frac{T - np}{p(1 - p)}, \quad \frac{\partial^2 \ell}{\partial p^2} = -\frac{T}{p^2} - \frac{n - T}{(1 - p)^2} < 0,$$

so ℓ is strictly concave and the first-order condition gives the unique maximizer $p = T/n$.

Step 5. *Conclusion.* $\hat{p}_{\text{ML}} = \bar{X} = \hat{p}_{\text{MM}}$.

The excluded cases are handled ad hoc: if $T = 0$ or $T = n$, then $L = (1 - p)^n$ or $L = p^n$ is monotone on $[0, 1]$, with maximum at $p = 0 = T/n$ or at $p = 1 = T/n$. So $\hat{p}_{\text{ML}} = \bar{X}$ for every sample.

Example 9.3 (Maximum likelihood: uniform). Let $X_i \sim \text{iid } U[0, \theta]$, $\Theta = (0, \infty)$; note that $X_{(1)} \geq 0$ with $\mathbb{P}_\theta$ -probability one.

Step 1. *The density.* $f(x \mid \theta) = \frac{1}{\theta} \mathbf{1}\{0 \leq x \leq \theta\}$. Its support depends on θ , so this is *not* an exponential family.

Step 2. *The likelihood.* Since all $X_i \leq \theta$ iff $X_{(n)} \leq \theta$, the data enter only through $X_{(n)}$ (Example 10.4):

$$L(\theta \mid X_1, \dots, X_n) = \prod_{i=1}^n \frac{1}{\theta} \mathbf{1}\{0 \leq X_i \leq \theta\} = \frac{1}{\theta^n} \mathbf{1}\{X_{(n)} \leq \theta\}.$$

Step 3. *The log likelihood.* $\ell = -n \log \theta$ for $\theta \geq X_{(n)}$, and $\ell = -\infty$ for $\theta < X_{(n)}$.

Step 4. *The maximum.* L is 0 for $\theta < X_{(n)}$ and strictly decreasing on $[X_{(n)}, \infty)$, so the maximizer is $\theta = X_{(n)}$. No first-order condition finds it: $\partial \ell / \partial \theta = -n/\theta$ never vanishes.

Step 5. *Conclusion.* $\hat{\theta}_{\text{ML}} = X_{(n)} \neq 2\bar{X} = \hat{\theta}_{\text{MM}}$. It never falls below an observation, but is biased downward: by Theorem 7.17, $X_{(n)}$ has pdf $nt^{n-1}\theta^{-n}$ on $(0, \theta)$, so $\mathbb{E}_\theta[X_{(n)}] = n\theta/(n+1) < \theta$.

Example 9.4 (Maximum likelihood: normal). Let $X_i \sim \text{iid } N(\mu, \sigma^2)$ with $n \geq 2$, $\theta = (\mu, \sigma^2) \in \Theta = \mathbb{R} \times (0, \infty)$, and write $\hat{\sigma}^2 = \frac{1}{n} \sum_i (X_i - \bar{X})^2$.

Step 1. *The density.* In the exponential-family form (59) of Example 5.5,

$$f(x \mid \theta) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) = \underbrace{\frac{e^{-\mu^2/(2\sigma^2)}}{\sqrt{2\pi} \sigma}}_{c(\theta)} \exp\left(\frac{\mu}{\sigma^2} x - \frac{1}{2\sigma^2} x^2\right).$$

Step 2. *The likelihood.*

$$L(\theta \mid X_1, \dots, X_n) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2\right).$$

Step 3. *The log likelihood.* By (84) with $a = \mu$, $\sum_i (X_i - \mu)^2 = n\hat{\sigma}^2 + n(\bar{X} - \mu)^2$, so

$$\ell(\mu, \sigma^2 \mid X_1, \dots, X_n) = -\frac{n}{2} \left[\log(2\pi\sigma^2) + \frac{\hat{\sigma}^2 + (\bar{X} - \mu)^2}{\sigma^2} \right].$$

Step 4. *The maximum.* For every $\sigma^2 > 0$, ℓ is strictly decreasing in $(\bar{X} - \mu)^2$, so it is maximized at $\mu = \bar{X}$. Setting $\mu = \bar{X}$,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2} \frac{\sigma^2 - \hat{\sigma}^2}{\sigma^4},$$

which is positive for $\sigma^2 < \hat{\sigma}^2$ and negative for $\sigma^2 > \hat{\sigma}^2$: the maximizer is $\sigma^2 = \hat{\sigma}^2$, which is positive with probability one because $n \geq 2$.

Step 5. *Conclusion.* $\hat{\mu}_{\text{ML}} = \bar{X}$ and $\hat{\sigma}_{\text{ML}}^2 = \hat{\sigma}^2 = \frac{n-1}{n} S^2$, again the method of moments estimators.

9.2 Evaluating point estimators

9.2.1 Mean squared error and unbiasedness

The subscript θ specifies the sampling law: $\mathbb{P}_\theta$ is probability when the parameter is θ , and $\mathbb{E}_\theta$ and Var_θ are expectation and variance under that law. For a statistic $W = W(\mathbf{X})$ and joint sample density f_θ ,

$$\mathbb{E}_\theta[W] = \int W(\mathbf{x})f_\theta(\mathbf{x}) d\mathbf{x}, \quad \text{Var}_\theta(W) = \mathbb{E}_\theta\left[(W - \mathbb{E}_\theta[W])^2\right].$$

Definition 9.5 (Mean squared error; bias). Let $\Theta \subseteq \mathbb{R}$, and let $\hat{\theta}$ be an estimator of θ with $\mathbb{E}_\theta[\hat{\theta}^2] < \infty$ for every $\theta \in \Theta$ (as are all estimators below). The **mean squared error** and the **bias** of $\hat{\theta}$ are the functions

$$\text{MSE}_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2], \quad \text{Bias}_\theta(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta, \quad \theta \in \Theta,$$

and $\hat{\theta}$ is **unbiased** iff $\text{Bias}_\theta(\hat{\theta}) = 0$ for every $\theta \in \Theta$.

Proposition 9.6 (Bias–variance decomposition). $\text{MSE}_\theta(\hat{\theta}) = \text{Bias}_\theta(\hat{\theta})^2 + \text{Var}_\theta(\hat{\theta})$ for every $\theta \in \Theta$.

Proof. Since θ is fixed, expand the square and use $\mathbb{E}_\theta[\hat{\theta}^2] = \text{Var}_\theta(\hat{\theta}) + (\mathbb{E}_\theta[\hat{\theta}])^2$:

$$\begin{aligned} \text{MSE}_\theta(\hat{\theta}) &= \mathbb{E}_\theta[\hat{\theta}^2] - 2\theta\mathbb{E}_\theta[\hat{\theta}] + \theta^2 \\ &= \text{Var}_\theta(\hat{\theta}) + (\mathbb{E}_\theta[\hat{\theta}] - \theta)^2 \\ &= \text{Var}_\theta(\hat{\theta}) + \text{Bias}_\theta(\hat{\theta})^2. \square \end{aligned}$$

Example 9.7 (Estimating a normal variance). Let $X_i \sim \text{iid } N(\mu, \sigma^2)$, $n \geq 2$, with estimand σ^2 . By Theorem 7.10(c) and Table 3, $\text{Var}_\theta(S^2) = 2\sigma^4/(n-1)$; S^2 is unbiased (Theorem 7.6) and $\hat{\sigma}_{\text{ML}}^2 = (n-1)S^2/n$ has bias $-\sigma^2/n$. By Proposition 9.6,

$$\text{MSE}_\theta(S^2) = \frac{2\sigma^4}{n-1} > \frac{(2n-1)\sigma^4}{n^2} = \frac{\sigma^4}{n^2} + \left(\frac{n-1}{n}\right)^2 \frac{2\sigma^4}{n-1} = \text{MSE}_\theta(\hat{\sigma}_{\text{ML}}^2),$$

since $(2n-1)(n-1) < 2n^2$: the biased estimator has smaller MSE for every (μ, σ^2) .

10 Lecture 10: Sufficiency and UMVU Estimation

10.1 UMVU estimators

Definition 10.1 (UMVU estimator). An unbiased estimator $\hat{\theta}$ of θ is **uniform minimum variance unbiased** (UMVU) iff $\text{Var}_{\theta}(\hat{\theta}) \leq \text{Var}_{\theta}(\tilde{\theta})$ for every $\theta \in \Theta$ and every unbiased estimator $\tilde{\theta}$ of θ .

“Uniform” means that the comparison holds at every θ . By Proposition 9.6, UMVU also means smallest MSE within the unbiased class. To see why this restriction matters, fix $\theta_0 \in \Theta$ and consider the constant estimator $\hat{\theta} \equiv \theta_0$:

$$\text{Var}_{\theta}(\hat{\theta}) = 0, \quad \text{Bias}_{\theta}(\hat{\theta}) = \theta_0 - \theta, \quad \text{MSE}_{\theta}(\hat{\theta}) = (\theta_0 - \theta)^2.$$

Its MSE is zero only when the chosen θ_0 equals the true parameter.

10.2 Sufficiency and the Rao–Blackwell theorem

Definition 10.2 (Sufficient statistic). A statistic $T = T(X_1, \dots, X_n)$ is **sufficient** for θ iff the conditional distribution of $(X_1, \dots, X_n)^{\top}$ given T does not depend on θ .⁷

So T carries all the information about θ in the sample [Casella and Berger, 2024, Thm. 6.2.6].

Theorem 10.3 (Factorization theorem; proof omitted). $T = T(X_1, \dots, X_n)$ is sufficient for θ iff there exist functions $g(\cdot | \theta) \geq 0$, $\theta \in \Theta$, and $h \geq 0$ such that, for every $\mathbf{x} \in \mathbb{R}^n$ and every $\theta \in \Theta$,

$$\prod_{i=1}^n f(x_i | \theta) = g(T(\mathbf{x}) | \theta) h(\mathbf{x}). \quad (94)$$

⁷Conditioning in the continuous case. Write $\mathbf{X} = (X_1, \dots, X_n)^{\top}$. In the discrete case we can use $\mathbb{P}_{\theta}(\mathbf{X} = \mathbf{x}, T = t) / \mathbb{P}_{\theta}(T = t)$ whenever the denominator is positive. For continuous T , $\mathbb{P}_{\theta}(T = t) = 0$, so this ratio is unavailable. Instead, a conditional distribution is a probability kernel $K_{\theta}(t, A)$: a probability law in A , measurable in t , satisfying

$$\mathbb{P}_{\theta}(\mathbf{X} \in A, T \in B) = \int_B K_{\theta}(t, A) \mathbb{P}_{\theta}^T(dt)$$

for all Borel sets A, B , where $\mathbb{P}_{\theta}^T$ is the law of T . It specifies the distribution of the sample given $T = t$, up to sets of t having probability zero. Sufficiency means that a single kernel $K(t, A)$ works for every θ . This definition also covers $(\mathbf{X}, T(\mathbf{X}))$, which generally has no joint Lebesgue density.

Example 10.4 (Uniform: $X_{(n)}$ is sufficient). For $X_i \sim \text{iid } U[0, \theta]$,

$$\prod_{i=1}^n \frac{1}{\theta} \mathbf{1}\{0 \leq x_i \leq \theta\} = \underbrace{\frac{1}{\theta^n} \mathbf{1}\{x_{(n)} \leq \theta\}}_{g(x_{(n)}|\theta)} \underbrace{\mathbf{1}\{x_{(1)} \geq 0\}}_{h(\mathbf{x})},$$

so $X_{(n)}$ is sufficient for θ by Theorem 10.3. Concretely, given $X_{(n)} = t > 0$, choose the index of the maximum uniformly from $\{1, \dots, n\}$, set that observation equal to t , and draw the other $n - 1$ observations independently from $U[0, t]$. This conditional law depends on t , but not on θ .

Conditional expectation. On a probability space $(S, \mathcal{F}, \mathbb{P})$, let $\mathbb{E}[|X|] < \infty$. Then $Z = \mathbb{E}[X | Y]$ is the integrable, $\sigma(Y)$ -measurable random variable satisfying

$$\int_A Z d\mathbb{P} = \int_A X d\mathbb{P} \quad \text{for every } A \in \sigma(Y).$$

Here $\sigma(Y)$ consists of the events determined by Y . Thus $Z = \varphi(Y)$ uses only Y and preserves the integral of X on each such event. It exists and is unique almost surely. When (X, Y) has a joint density, this agrees with Definition 6.3:

$$\varphi(y) = \mathbb{E}[X | Y = y] = \int x \frac{f_{X,Y}(x, y)}{f_Y(y)} dx \quad \text{for almost every } y \text{ with } f_Y(y) > 0.$$

Thus $\mathbb{E}[X | Y = y]$ is a number, while $\mathbb{E}[X | Y]$ is a random variable.

Law of iterated expectations. If $\mathbb{E}[|X|] < \infty$, then $\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$.

Proof. Let $Z = \mathbb{E}[X | Y]$. The whole sample space $S = \{Y \in \mathbb{R}\}$ is an event determined by Y , so $S \in \sigma(Y)$. Taking $A = S$ in the defining identity,

$$\mathbb{E}[\mathbb{E}[X | Y]] = \int_S Z d\mathbb{P} = \int_S X d\mathbb{P} = \mathbb{E}[X],$$

where both integrals are finite because Z and X are integrable. $\square$

Theorem 10.5 (Rao–Blackwell). *Let $\tilde{\theta}$ be an unbiased estimator of θ with finite variance and let T be sufficient for θ . Then $\hat{\theta} = \mathbb{E}_{\theta}[\tilde{\theta} | T]$ is an unbiased estimator of θ with $\text{Var}_{\theta}(\hat{\theta}) \leq \text{Var}_{\theta}(\tilde{\theta})$ for every $\theta \in \Theta$, with equality iff $\mathbb{P}_{\theta}(\hat{\theta} = \tilde{\theta}) = 1$.*

Proof. By sufficiency the conditional distribution of $\tilde{\theta}$ given T does not depend on θ , so $\hat{\theta} = \varphi(T)$ with φ free of θ : $\hat{\theta}$ is a statistic. Under $\mathbb{P}_{\theta}$, the law of iterated expectations

(Theorem 6.9) gives $\mathbb{E}_\theta[\hat{\theta}] = \mathbb{E}_\theta[\mathbb{E}_\theta[\tilde{\theta} | T]] = \mathbb{E}_\theta[\tilde{\theta}] = \theta$. The conditional variance identity (Theorem 6.10) gives

$$\begin{aligned}\text{Var}_\theta(\tilde{\theta}) &= \text{Var}_\theta(\hat{\theta}) + \mathbb{E}_\theta[\text{Var}_\theta(\tilde{\theta} | T)] \\ &= \text{Var}_\theta(\hat{\theta}) + \mathbb{E}_\theta[(\tilde{\theta} - \hat{\theta})^2] \geq \text{Var}_\theta(\hat{\theta}).\end{aligned}$$

Equality holds iff the nonnegative squared difference has expectation zero, equivalently $\mathbb{P}_\theta(\hat{\theta} = \tilde{\theta}) = 1$. $\square$

So the search for UMVU estimators can be restricted to functions of a sufficient statistic. In the $U[0, \theta]$ model, $\mathbb{E}_\theta[2\bar{X} | X_{(n)}]$ improves on $\hat{\theta}_{\text{MM}} = 2\bar{X}$.

10.3 Completeness and Lehmann–Scheffé

If $X \sim \text{Bernoulli}(p)$ and $T = X$, then $\mathbb{P}_p(T = 1) = p$ and

$$\mathbb{E}_p[g(T)] = (1 - p)g(0) + pg(1).$$

The function g stays the same as p varies; only the probability weights change. This expectation can vanish at one p by cancellation: $g(t) = t - 1/2$ has mean zero at $p = 1/2$. Requiring it to vanish for *every* $p \in (0, 1)$ forces $g(0) = g(1) = 0$. Completeness formalizes this distinction.

Definition 10.6 (Complete statistic). A statistic T is **complete** iff, for every Borel function g that does not depend on θ and satisfies $\mathbb{E}_\theta[|g(T)|] < \infty$ for every $\theta \in \Theta$,

$$\mathbb{E}_\theta[g(T)] = 0 \text{ for every } \theta \in \Theta \implies \mathbb{P}_\theta(g(T) = 0) = 1 \text{ for every } \theta \in \Theta.$$

Here $\mathbb{P}_\theta(g(T) = 0) = 1$ means that $g(T)$ equals zero almost surely under that sampling law, a stronger statement than having mean zero. Completeness rules out a function of T that is nonzero with positive probability under some θ , yet has mean zero under *every* θ in the family.

Theorem 10.7 (Lehmann–Scheffé).

- (a) If T is complete and $\varphi(T)$ and $\psi(T)$ are both unbiased for θ , then $\mathbb{P}_\theta(\varphi(T) = \psi(T)) = 1$ for every $\theta \in \Theta$.
- (b) If T is sufficient and complete and $\hat{\theta} = \varphi(T)$ is unbiased for θ , then $\hat{\theta}$ is UMVU.

Proof. (a) Apply Definition 10.6 to $g = \varphi - \psi$, for which $\mathbb{E}_\theta[g(T)] = \theta - \theta = 0$. (b) For any unbiased $\tilde{\theta}$, $\mathbb{E}_\theta[\tilde{\theta} \mid T]$ is an unbiased function of T (Theorem 10.5), so it equals $\hat{\theta}$ with probability one by (a), and $\text{Var}_\theta(\hat{\theta}) \leq \text{Var}_\theta(\tilde{\theta})$. $\square$

Example 10.8 (Uniform: the UMVU estimator). Let $X_i \sim \text{iid } U[0, \theta]$ and $T = X_{(n)}$, sufficient by Example 10.4. By Theorem 7.17, T has pdf $nt^{n-1}\theta^{-n}$ on $(0, \theta)$, so $\mathbb{E}_\theta[g(T)] = 0$ for every $\theta > 0$ means $\int_0^\theta g(t) t^{n-1} dt = 0$ for every $\theta > 0$. Differentiating in θ , $g(\theta) = 0$ for (almost) every $\theta > 0$: T is complete [Casella and Berger, 2024, Ex. 6.2.23]. Since $\mathbb{E}_\theta[X_{(n)}] = \int_0^\theta t nt^{n-1}\theta^{-n} dt = n\theta/(n+1)$,

$$\hat{\theta} = \frac{n+1}{n} X_{(n)}$$

is UMVU by Theorem 10.7(b), and $\mathbb{E}_\theta[2\bar{X} \mid X_{(n)}] = \hat{\theta}$ by Theorem 10.7(a). Moreover $\mathbb{E}_\theta[X_{(n)}^2] = n\theta^2/(n+2)$, so $\text{Var}_\theta(X_{(n)}) = n\theta^2/[(n+1)^2(n+2)]$, while $\text{Var}_\theta(2\bar{X}) = 4\theta^2/(12n)$ by Theorem 7.6(b) and Table 3:

$$\text{Var}_\theta\left(\frac{n+1}{n}X_{(n)}\right) = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{3n} = \text{Var}_\theta(2\bar{X}) \quad (n \geq 2).$$

Theorem 10.9 (Exponential families). *Let $X_1, \dots, X_n$ be a random sample from the exponential family (59), written as*

$$f(x \mid \theta) = h(x)c(\theta) \exp\left(\sum_{j=1}^k w_j(\theta)t_j(x)\right),$$

where $t_1, \dots, t_k$ are fixed functions of a single observation and do not depend on θ . Define

$$T_j = \sum_{i=1}^n t_j(X_i), \quad j = 1, \dots, k, \quad \mathbf{T} = (T_1, \dots, T_k)^\top.$$

Here i indexes the observations and j indexes the functions: $t_j(X_i)$ is the j th function evaluated at the i th observation. Then $\mathbf{T}$ is sufficient for θ , and it is complete if $\{(w_1(\theta), \dots, w_k(\theta))^\top : \theta \in \Theta\}$ contains an open subset of $\mathbb{R}^k$.

Proof. Sufficiency: $\prod_i f(x_i \mid \theta) = c(\theta)^n \exp(\sum_j w_j(\theta) T_j(\mathbf{x})) \cdot \prod_i h(x_i)$ and Theorem 10.3. For completeness, see Casella and Berger [2024, Thm. 6.2.25]. $\square$

Example 10.10 (Bernoulli and normal: UMVU estimators).

- (1) $\text{Bernoulli}(p)$, $0 < p < 1$. By Example 5.4, $t(x) = x$ and $w(p) = \log \frac{p}{1-p}$ ranges over $\mathbb{R}$. So $T = \sum_i X_i$ is complete and sufficient, and the unbiased $\bar{X} = T/n$ is UMVU: $\hat{p}_{\text{ML}} = \hat{p}_{\text{MM}} = \hat{p}_{\text{UMVU}}$.
- (2) $N(\mu, \sigma^2)$. By Example 5.5, $\mathbf{t}(x) = (x, x^2)^\top$ and $\mathbf{w}(\theta) = (\mu/\sigma^2, -1/(2\sigma^2))^\top$ ranges over the open set $\mathbb{R} \times (-\infty, 0)$. So $\mathbf{T} = (\sum_i X_i, \sum_i X_i^2)^\top$ is complete and sufficient, and by Theorem 7.4(b) $\bar{X} = T_1/n$ and $S^2 = (T_2 - T_1^2/n)/(n-1)$ are unbiased functions of $\mathbf{T}$, hence UMVU for μ and σ^2 ; yet $\hat{\sigma}_{\text{ML}}^2$ has smaller MSE (Example 9.7).

10.4 The Cramér–Rao inequality

The precision of an unbiased estimator depends on how well the data distinguish parameter values. If nearby values of θ generate very similar distributions, the data have little ability to distinguish them, so an estimator cannot concentrate too tightly around each possible value. The Cramér–Rao inequality makes this tradeoff precise: it gives a lower bound on sampling variance in terms of how sensitive the distribution is to changes in θ .

Write $L_\theta(\mathbf{x}) = \prod_{i=1}^n f(x_i | \theta)$. The individual and sample **scores** are the slopes of the log density and log likelihood, with the data held fixed when differentiating:

$$s(x, \theta) = \frac{\partial}{\partial \theta} \log f(x | \theta), \quad S_\theta = S_\theta(\mathbf{X}) = \frac{\partial}{\partial \theta} \log L_\theta(\mathbf{X}) = \sum_{i=1}^n s(X_i, \theta).$$

A large absolute score means that a small change in θ substantially changes the relative likelihood of those data. **Fisher information** averages the squared score over possible data generated at the fixed value θ . Under the regularity conditions below,

$$I_1(\theta) = \mathbb{E}_\theta[s(X_1, \theta)^2], \quad I(\theta) = \mathbb{E}_\theta[S_\theta^2] = nI_1(\theta).$$

Thus $I_1(\theta)$ is information from *one observation*, whereas $I(\theta)$ is information from the *whole sample*. These are functions of the model and θ , not random quantities computed from one realized sample. Independent observations contribute additively to information.

Theorem 10.11 (Cramér–Rao inequality). *Let $X_1, \dots, X_n$ be a random sample from $f(\cdot \mid \theta)$, where $\Theta \subseteq \mathbb{R}$ is an open interval, and let $\hat{\theta} = W(X_1, \dots, X_n)$ have finite variance and satisfy $\mathbb{E}_\theta[\hat{\theta}] = \theta$ for every $\theta \in \Theta$. Suppose that:*

(R1) *the support $\mathcal{X} = \{x : f(x \mid \theta) > 0\}$ does not depend on θ , and $\theta \mapsto f(x \mid \theta)$ is differentiable on Θ for every $x \in \mathcal{X}$;*

(R2) *differentiation with respect to θ can be taken under the integral sign (the sum, in the discrete case) in $\int f(x \mid \theta) dx = 1$ and in $\int W(\mathbf{x}) \prod_i f(x_i \mid \theta) d\mathbf{x} = \theta$;*

(R3) $0 < I_1(\theta) < \infty$.

Then

$$\text{Var}_\theta(\hat{\theta}) \geq \frac{1}{I(\theta)} \quad \text{for every } \theta \in \Theta. \quad (95)$$

Proof. First, compute the mean and variance of the score. By (R1)–(R2),

$$\begin{aligned} \mathbb{E}_\theta[s(X_1, \theta)] &= \int_{\mathcal{X}} \frac{\partial_\theta f(x \mid \theta)}{f(x \mid \theta)} f(x \mid \theta) dx \\ &= \frac{d}{d\theta} \int_{\mathcal{X}} f(x \mid \theta) dx = \frac{d}{d\theta} 1 = 0. \end{aligned}$$

Thus $\mathbb{E}_\theta[S_\theta] = 0$. Independence of the observations gives

$$\text{Var}_\theta(S_\theta) = \sum_{i=1}^n \text{Var}_\theta(s(X_i, \theta)) = nI_1(\theta) = I(\theta).$$

Next, relate the score to the estimator. Since $\partial_\theta L_\theta(\mathbf{x}) = S_\theta(\mathbf{x})L_\theta(\mathbf{x})$, differentiating the unbiasedness identity under (R2) yields

$$1 = \frac{d}{d\theta} \mathbb{E}_\theta[\hat{\theta}] = \int W(\mathbf{x}) S_\theta(\mathbf{x}) L_\theta(\mathbf{x}) d\mathbf{x} = \mathbb{E}_\theta[\hat{\theta} S_\theta].$$

Because $\mathbb{E}_\theta[S_\theta] = 0$, this means $\text{Cov}_\theta(\hat{\theta}, S_\theta) = 1$.

Finally, apply Cauchy–Schwarz (Theorem B.3):

$$1 = \text{Cov}_\theta(\hat{\theta}, S_\theta)^2 \leq \text{Var}_\theta(\hat{\theta}) \text{Var}_\theta(S_\theta) = \text{Var}_\theta(\hat{\theta}) I(\theta).$$

Divide by $I(\theta) > 0$. The discrete case uses sums in place of integrals. □

With twice differentiable densities and justification for differentiating $\int f(x | \theta) dx = 1$ twice, an equivalent formula is

$$I_1(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X_1 | \theta) \right].$$

This expresses information as the expected downward curvature of the log likelihood: a sharper likelihood separates nearby parameter values more strongly [Casella and Berger, 2024, Sec. 7.3.2].

Example 10.12 (Three possibilities).

(1) *Attained.* For Bernoulli(p) with $0 < p < 1$, the sample mean attains the bound:

$$\begin{aligned} s(x, p) &= \frac{x - p}{p(1 - p)}, & I_1(p) &= \frac{1}{p(1 - p)}, \\ \text{Var}_p(\bar{X}) &= \frac{p(1 - p)}{n} = \frac{1}{I(p)}. \end{aligned}$$

(2) *Not attained by S^2 .* For σ^2 in $N(\mu, \sigma^2)$ with μ known and $n \geq 2$, the information is $I_1(\sigma^2) = 1/(2\sigma^4)$. We compute the variance of $S^2 = (n - 1)^{-1} \sum_{i=1}^n (X_i - \bar{X})^2$ directly from moments. Let $Y_i = X_i - \mu$, so the Y_i are independent $N(0, \sigma^2)$ variables. Their mgf is $M_Y(t) = \exp(\sigma^2 t^2/2)$, whose derivatives at zero give

$$\mathbb{E}[Y_i] = 0, \quad \mathbb{E}[Y_i^2] = \sigma^2, \quad \mathbb{E}[Y_i^4] = 3\sigma^4.$$

Define $A = \sum_{i=1}^n Y_i^2$ and $B = \sum_{1 \leq i < j \leq n} Y_i Y_j$. Expanding the squared sample mean,

$$\begin{aligned} (n - 1)S^2 &= \sum_{i=1}^n Y_i^2 - \frac{1}{n} \left(\sum_{i=1}^n Y_i \right)^2 = \frac{n - 1}{n} A - \frac{2}{n} B, \\ S^2 &= \frac{A}{n} - \frac{2B}{n(n - 1)}. \end{aligned}$$

Independence gives $\text{Var}(A) = n[\mathbb{E}[Y_i^4] - (\mathbb{E}[Y_i^2])^2] = 2n\sigma^4$. For $i < j$, $\mathbb{E}[Y_i Y_j] = 0$ and $\text{Var}(Y_i Y_j) = \mathbb{E}[Y_i^2] \mathbb{E}[Y_j^2] = \sigma^4$. Any two distinct pair products are uncorrelated: if their index pairs are disjoint, independence makes the expectation of their product zero; if they share one index, for example (i, j) and (i, k) , then $\mathbb{E}[Y_i^2 Y_j Y_k] = \mathbb{E}[Y_i^2] \mathbb{E}[Y_j] \mathbb{E}[Y_k] = 0$. There are $\binom{n}{2}$ pairs, hence $\text{Var}(B) = \binom{n}{2} \sigma^4$.

Also $\text{Cov}(A, B) = 0$: each term $\mathbb{E}[Y_i^2 Y_j Y_k]$ with $j < k$ contains at least one independent, centered factor to the first power, even when $i = j$ or $i = k$. Therefore

$$\begin{aligned}\text{Var}(S^2) &= \frac{\text{Var}(A)}{n^2} + \frac{4 \text{Var}(B)}{n^2(n-1)^2} - \frac{4 \text{Cov}(A, B)}{n^2(n-1)} \\ &= \frac{2\sigma^4}{n} + \frac{2\sigma^4}{n(n-1)} = \boxed{\frac{2\sigma^4}{n-1}}.\end{aligned}$$

Thus $\text{Var}(S^2) = 2\sigma^4/(n-1) > 2\sigma^4/n = 1/I(\sigma^2)$. With known μ , $n^{-1} \sum_i (X_i - \mu)^2$ is unbiased and attains the bound, so S^2 is not UMVU.

- (3) *Not applicable.* For $U[0, \theta]$, (R1) fails because the support depends on θ : differentiating an integral with upper limit θ contributes a boundary term. Indeed, the interior score is $s(x, \theta) = -1/\theta$, whose expectation is not zero, so the proof breaks down. The formal “bound” θ^2/n exceeds the actual UMVU variance:

$$\text{Var}_\theta \left(\frac{n+1}{n} X_{(n)} \right) = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{n} \quad (\text{Example 10.8}).$$

11 Lecture 11: Hypothesis Testing

11.1 The testing problem

The setup is that of Section 8.2: $X_1, \dots, X_n$ is a random sample from $f(\cdot \mid \theta)$, with $\theta \in \Theta$ unknown. The objective is now to choose between two subsets of Θ .

Definition 11.1 (Hypotheses). A **hypothesis** is a statement about θ . Partition Θ into two disjoint subsets, Θ_0 and $\Theta_1 = \Theta \setminus \Theta_0 = \{\theta \in \Theta : \theta \notin \Theta_0\}$. The **null hypothesis** is $H_0 : \theta \in \Theta_0$, the **alternative hypothesis** is $H_1 : \theta \in \Theta_1$, and the **maintained hypothesis** is $H : \theta \in \Theta$. A hypothesis is **simple** if the corresponding subset is a singleton, and **composite** otherwise.

Example 11.2 (Two formulations). Let $X_i \sim \text{iid } N(\mu, 1)$ with $\mu \geq 0$ unknown, and suppose the aim is to test whether $\mu = 0$. Both

$$(1) H_0 : \mu = 0 \text{ vs. } H_1 : \mu > 0, \quad (2) H_0 : \mu > 0 \text{ vs. } H_1 : \mu = 0$$

partition $\Theta = [0, \infty)$. Formulation (1) is the usual one: the statement to be disproved by the data is made the null.

Definition 11.3 (Testing procedure; critical region). A **testing procedure** (a **test**) is a data-based rule for choosing between H_0 and H_1 . For $C \subseteq \mathbb{R}^n$, the rule “reject H_0 iff $(X_1, \dots, X_n)^\top \in C$ ” is a testing procedure with **critical region** C .

Every critical region can be written as $C = \{\mathbf{x} \in \mathbb{R}^n : T(\mathbf{x}) > c\}$ for some $T : \mathbb{R}^n \rightarrow \mathbb{R}$ and some $c \in \mathbb{R}$: take $T = \mathbf{1}\{\mathbf{x} \in C\}$ and $c = 1/2$. Then $T(X_1, \dots, X_n)$ is the **test statistic** and c is the **critical value**, so choosing a test means choosing these two objects.

Example 11.4 (A one-sided normal test). Let $X_i \sim \text{iid } N(\mu, 1)$ and consider $H_0 : \mu = 0$ vs. $H_1 : \mu > 0$. The rule “reject H_0 iff $\bar{X} > 1.645/\sqrt{n}$ ” has test statistic $\sqrt{n} \bar{X}$, critical value 1.645, and critical region $C = \{\mathbf{x} \in \mathbb{R}^n : \sqrt{n} \bar{x} > 1.645\}$.

11.2 The power function

A test can err in two ways: it can reject H_0 when H_0 is true (a **type I error**, or **false positive**) or fail to reject H_0 when H_1 is true (a **type II error**, or **false negative**).

Here “positive” means rejecting H_0 in favor of H_1 ; failure to reject does not establish that H_0 is true.

Table 6: The two errors of a test.

	Fail to reject H_0	Reject H_0
H_0 true	correct	type I error (false positive)
H_1 true	type II error (false negative)	correct

Definition 11.5 (Power function; size; level). The **power function** of a test with critical region $C \subseteq \mathbb{R}^n$ is the function $\beta : \Theta \rightarrow [0, 1]$ given by

$$\beta(\theta) = \mathbb{P}_\theta((X_1, \dots, X_n)^\top \in C) = \mathbb{P}_\theta(\text{reject } H_0), \quad \theta \in \Theta. \quad (96)$$

Its **size** is the supremum of rejection probabilities under the null, and it is of **level** α if that size is at most α :

$$\text{size}(C) := \sup_{\theta \in \Theta_0} \beta(\theta) = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\text{reject } H_0), \quad (97)$$

$$C \text{ is a level } \alpha \text{ test} \iff \text{size}(C) \leq \alpha. \quad (98)$$

The power function collects both error probabilities: $\mathbb{P}_\theta(\text{type I error}) = \beta(\theta)$ for $\theta \in \Theta_0$, and $\mathbb{P}_\theta(\text{type II error}) = 1 - \beta(\theta)$ for $\theta \in \Theta_1$. The **ideal power function** makes neither error:

$$\beta_{\text{ideal}}(\theta) = \begin{cases} 0, & \theta \in \Theta_0, \\ 1, & \theta \in \Theta_1. \end{cases} \quad (99)$$

When the null and alternative sampling distributions have strictly positive densities on the same support, as in the normal example, no finite-sample test achieves this ideal. For a rejection rule $T > c$, lowering c increases the probability of a type I error and decreases that of a type II error (weakly in each case). The standard resolution is asymmetric: cap the type I error at a small level α (conventionally 5%) and, subject to that, make β as large as possible on Θ_1 (Section 12.1).

Example 11.6 (Power of the one-sided normal test). Continuing Example 11.4: under $\mathbb{P}_\mu$, $\sqrt{n}(\bar{X} - \mu) \sim N(0, 1)$ by Theorem 7.10(b), so

$$\beta(\mu) = \mathbb{P}_\mu(\sqrt{n}\bar{X} > 1.645) = \mathbb{P}_\mu(\sqrt{n}(\bar{X} - \mu) > 1.645 - \sqrt{n}\mu) = 1 - \Phi(1.645 - \sqrt{n}\mu),$$

where Φ is the standard normal cdf. Thus β is strictly increasing, the size is $\beta(0) = 1 - \Phi(1.645) \approx 0.05$, and $\beta(\mu) \rightarrow 1$ as $\mu \rightarrow \infty$ and as $n \rightarrow \infty$ for fixed $\mu > 0$.

12 Lecture 12: Optimal Tests

12.1 Uniformly most powerful tests

Definition 12.1 (UMP test). A level α test with power function $\beta(\cdot)$ is a **uniformly most powerful** (UMP) level α test iff

$$\beta(\theta) \geq \tilde{\beta}(\theta) \quad \text{for every } \theta \in \Theta_1 \quad (100)$$

and every power function $\tilde{\beta}(\cdot)$ of a level α test.

“Uniformly” again refers to the quantifier over θ , and the comparison is restricted to level α tests, exactly as UMVU restricts the comparison to unbiased estimators (Definition 10.1). Whether a UMP test exists depends on the alternative. For $H_0 : \theta = \theta_0$ with $\theta_0 \in \mathbb{R}$:

- (1) a UMP level α test *always* exists if H_1 is simple (Theorem 12.2);
- (2) one *often* exists if $H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$ (Section 12.2);
- (3) one often *fails* to exist if $H_1 : \theta \neq \theta_0$; a UMP *unbiased* test often does (Section 12.3).

Theorem 12.2 (Neyman–Pearson). Consider $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$, and let $L(\cdot | X_1, \dots, X_n)$ be the likelihood (93). For $k \geq 0$, the test that rejects H_0 iff

$$L(\theta_1 | X_1, \dots, X_n) > k L(\theta_0 | X_1, \dots, X_n) \quad (101)$$

is a UMP level α test, where $\alpha = \mathbb{P}_{\theta_0}(L(\theta_1 | X_1, \dots, X_n) > k L(\theta_0 | X_1, \dots, X_n))$. Conversely, every UMP level α test is of this form, except possibly on an event of probability zero under $\mathbb{P}_{\theta_0}$ and $\mathbb{P}_{\theta_1}$.

Where $L(\theta_0 | X_1, \dots, X_n) > 0$, (101) says that the **likelihood ratio** exceeds k ,

$$\frac{L(\theta_1 | X_1, \dots, X_n)}{L(\theta_0 | X_1, \dots, X_n)} > k :$$

reject when the data are sufficiently more likely under H_1 than under H_0 . As k ranges over $[0, \infty)$ the attainable sizes form a set that need not contain a prescribed α ; in the

discrete case this is repaired by randomizing on the boundary $\{L(\theta_1 | \cdot) = k L(\theta_0 | \cdot)\}$ [Casella and Berger, 2024, §8.3.2].

Example 12.3 (Normal mean: the Neyman–Pearson test). Let $X_i \sim \text{iid } N(\mu, 1)$ and test $H_0 : \mu = 0$ vs. $H_1 : \mu = \mu_1$ with $\mu_1 > 0$. Expanding the square in (93),

$$L(\mu | X_1, \dots, X_n) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n X_i^2\right) \exp\left(-\frac{n}{2}\mu^2 + \mu \sum_{i=1}^n X_i\right),$$

so the likelihood ratio is

$$\frac{L(\mu_1 | X_1, \dots, X_n)}{L(0 | X_1, \dots, X_n)} = \exp\left(-\frac{n}{2}\mu_1^2 + n\mu_1 \bar{X}\right),$$

strictly increasing in $\bar{X}$ because $\mu_1 > 0$. Hence (101) is equivalent to $\bar{X} > c$ with $c = \frac{\log k}{n\mu_1} + \frac{\mu_1}{2}$, and by Example 11.6 the choice $c = 1.645/\sqrt{n}$ gives size 0.05. The UMP level .05 test rejects H_0 iff $\bar{X} > 1.645/\sqrt{n}$.

12.2 One-sided alternatives

The test of Example 12.3 does not depend on μ_1 , and that is the whole point. In general, for $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$:

- (1) for each $\theta_1 > \theta_0$, Theorem 12.2 gives the UMP level α test of $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$;
- (2) if these tests coincide for every $\theta_1 > \theta_0$, the common test is UMP level α against $H_1 : \theta > \theta_0$, since its power is maximal at each θ_1 separately;
- (3) otherwise no UMP level α test exists.

Example 12.4 (Normal mean, continued). For $X_i \sim \text{iid } N(\mu, 1)$, Example 12.3 holds for every $\mu_1 > 0$, so the UMP level .05 test of $H_0 : \mu = 0$ vs. $H_1 : \mu > 0$ rejects iff $\bar{X} > 1.645/\sqrt{n}$. Running the same argument with $\mu_1 < 0$ reverses the inequality: the UMP level .05 test against $H_1 : \mu < 0$ rejects iff $\bar{X} < -1.645/\sqrt{n}$. A UMP level .05 test against $H_1 : \mu \neq 0$ would have to agree with both, so none exists.

The pattern in Example 12.3 — a likelihood ratio that is monotone in a statistic — is what makes step (2) work.

12.3 Unbiased tests

Definition 12.5 (Unbiased test). A test of $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ with power function $\beta(\cdot)$ is **unbiased** iff

$$\sup_{\theta \in \Theta_0} \beta(\theta) \leq \inf_{\theta \in \Theta_1} \beta(\theta). \quad (102)$$

A **UMP unbiased** level α test is one that is UMP within the class of unbiased level α tests.

In words, the test is never more likely to reject when H_1 holds than when H_0 holds. For a simple null $H_0 : \theta = \theta_0$ with a differentiable power function, (102) says that β is minimized at θ_0 , so an unbiased level α test satisfies the two constraints

$$\beta(\theta_0) = \alpha, \quad \beta'(\theta_0) = 0.$$

Maximizing power subject to two constraints is what the generalized Neyman–Pearson lemma does [Casella and Berger, 2024, §8.3.3].

Example 12.6 (Two-sided normal test). Let $X_i \sim \text{iid } N(\mu, 1)$ and test $H_0 : \mu = 0$ vs. $H_1 : \mu \neq 0$, for which no UMP test exists (Example 12.4). The UMP unbiased level .05 test rejects H_0 iff $|\bar{X}| > 1.96/\sqrt{n}$. Its power function is

$$\beta(\mu) = 1 - \Phi(1.96 - \sqrt{n}\mu) + \Phi(-1.96 - \sqrt{n}\mu),$$

and indeed $\beta(0) = 0.05$ and $\beta'(0) = \sqrt{n} [\varphi(1.96) - \varphi(-1.96)] = 0$, where $\varphi = \Phi'$.

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A The Gamma and Beta Functions

Reading: Casella and Berger [2024, §3.3], which also has the proofs omitted here.

Definition A.1 (Gamma and beta functions). For $\alpha, \beta > 0$,

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt, \quad B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx. \quad (103)$$

Proposition A.2 (Functional equation). For $\alpha > 0$ and $n \in \mathbb{N}$,

$$\Gamma(\alpha + 1) = \alpha \Gamma(\alpha), \quad \Gamma(n) = (n-1)!, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}. \quad (104)$$

Proof. Integrate (103) by parts with $u = t^\alpha$ and $dv = e^{-t} dt$; since $\Gamma(1) = 1$, induction gives the factorial. For $\Gamma(1/2)$ substitute $t = z^2/2$ and use the Gaussian integral (54). $\square$

Proposition A.3 (Beta in terms of gamma; proof omitted). $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$.
Consequently

$$B(\alpha, \beta) = B(\beta, \alpha), \quad B(1, 1) = 1, \quad \frac{B(\alpha + 1, \beta)}{B(\alpha, \beta)} = \frac{\alpha}{\alpha + \beta}. \quad (105)$$

Write $\Gamma(\alpha)\Gamma(\beta)$ as a double integral and change variables to $(u, v) = (s+t, s/(s+t))$, whose Jacobian is u ; the identities in (105) then follow from (104).

B Covariance, Correlation, and Normal Vectors

Reading: Casella and Berger [2024, §§4.3, 4.5–4.6], where the proofs omitted here can be found.

B.1 Covariance and correlation

Definition B.1 (Covariance, correlation).

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \quad (106)$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var } X \text{ Var } Y}}. \quad (107)$$

Theorem B.2. $\text{Var}(aX + bY) = a^2 \text{Var } X + b^2 \text{Var } Y + 2ab \text{Cov}(X, Y)$. If X and Y are independent then $\text{Cov}(X, Y) = 0$ and variances add; the converse is false.

Proof. Expand $\text{Var}(aX + bY) = \mathbb{E} \left[[a(X - \mathbb{E}[X]) + b(Y - \mathbb{E}[Y])]^2 \right]$ and use linearity; independence gives $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ by (77). For the converse take $X \sim N(0, 1)$ and $Y = X^2$: $\text{Cov}(X, Y) = \mathbb{E}[X^3] = 0$, yet Y is a function of X . $\square$

Theorem B.3 (Cauchy–Schwarz). $(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2]$, with equality if and only if $\mathbb{P}(Y = cX) = 1$ for some constant c . Consequently $|\rho_{XY}| \leq 1$, with $|\rho_{XY}| = 1$ if and only if Y is an affine function of X almost surely.

Proof. For every $t \in \mathbb{R}$, $q(t) = \mathbb{E}[(Y - tX)^2] = \mathbb{E}[Y^2] - 2t\mathbb{E}[XY] + t^2\mathbb{E}[X^2] \geq 0$, so this quadratic has nonpositive discriminant, which is the inequality. Equality means $q(t^*) = 0$ for some t^* , that is $Y = t^*X$ almost surely. Applying this to $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$ gives the statement about ρ . $\square$

B.2 Bivariate transformations

Theorem B.4 (Change of variables in $\mathbb{R}^2$; proof omitted). Let (X, Y) have density $f_{X,Y}$ on $\mathcal{A} \subset \mathbb{R}^2$ and let $(U, V) = (g_1(X, Y), g_2(X, Y))$ be a bijection from $\mathcal{A}$ onto $\mathcal{B}$ with continuously differentiable inverse $(x, y) = (h_1(u, v), h_2(u, v))$. Then

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) |J(u, v)|, \quad J = \det \begin{pmatrix} \partial h_1 / \partial u & \partial h_1 / \partial v \\ \partial h_2 / \partial u & \partial h_2 / \partial v \end{pmatrix}, \quad (108)$$

for $(u, v) \in \mathcal{B}$ and 0 elsewhere. If the map is a bijection only piecewise, sum (108) over the pieces, as in Theorem 2.23.

This is the change-of-variables theorem for double integrals applied to $\mathbb{P}((U, V) \in C)$; J is the Jacobian of the inverse, and for $n = 1$ it reduces to the $\left| \frac{d}{dy} g^{-1}(y) \right|$ of Theorem 2.22.

Example B.5 (Sum of independent variables). Take $U = X + Y$, $V = Y$, so $x = u - v$, $y = v$ and $J = 1$. Then $f_{U,V}(u, v) = f_{X,Y}(u - v, v)$, and marginalizing gives the **convolution formula**

$$f_{X+Y}(u) = \int_{-\infty}^{\infty} f_{X,Y}(u - v, v) dv = \int_{-\infty}^{\infty} f_X(u - v) f_Y(v) dv, \quad (109)$$

the second equality under independence.

B.3 The bivariate normal

Definition B.6 (Bivariate normal). (X, Y) is **bivariate normal** with parameters $\mu_X, \mu_Y \in \mathbb{R}$, $\sigma_X, \sigma_Y > 0$, $|\rho| < 1$ if

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} [z_X^2 - 2\rho z_X z_Y + z_Y^2]\right), \quad (110)$$

where $z_X = (x - \mu_X)/\sigma_X$ and $z_Y = (y - \mu_Y)/\sigma_Y$.

Theorem B.7. For (110):

- (a) $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$;
- (b) $\text{Cor}(X, Y) = \rho$;
- (c) every affine combination $aX + bY$ is normal;
- (d) the conditional distribution is normal, with

$$Y \mid X = x \sim N\left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2)\right); \quad (111)$$

- (e) X and Y are independent if and only if $\rho = 0$.

Proof. Completing the square in z_Y factors (110) as $f(x, y) = f_X(x) f(y \mid x)$, with f_X the $N(\mu_X, \sigma_X^2)$ density and $f(\cdot \mid x)$ the density in (111); this gives (a) and (d), and (b) follows from (111) and Theorem 6.9. Part (c) is a computation with the mgf, and in (e) the exponent separates exactly when $\rho = 0$, the converse being Theorem B.2. $\square$

Part (e) is special to the normal. Part (d) is the regression line: the conditional mean is affine in x with slope $\rho\sigma_Y/\sigma_X = \text{Cov}(X, Y)/\text{Var } X$, the population least-squares coefficient, and the conditional variance $\sigma_Y^2(1 - \rho^2)$ does not depend on x , so homoskedasticity holds by construction.

B.4 The multivariate normal

Definition B.8 (Multivariate normal). A random vector $\mathbf{X} = (X_1, \dots, X_n)^\top$ is **multivariate normal**, written $\mathbf{X} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with $\boldsymbol{\mu} \in \mathbb{R}^n$ and $\boldsymbol{\Sigma}$ symmetric positive definite, if

$$f(\mathbf{x}) = (2\pi)^{-n/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} \in \mathbb{R}^n. \quad (112)$$

With $n = 2$ and $\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$, (112) is (110).

Theorem B.9 (Multivariate normal mgf). *If $\mathbf{X} \sim N_n(\boldsymbol{\mu}, \Sigma)$ then*

$$M_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}\left[e^{\mathbf{t}^\top \mathbf{X}}\right] = \exp(\mathbf{t}^\top \boldsymbol{\mu} + \tfrac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t}), \quad \mathbf{t} \in \mathbb{R}^n, \quad (113)$$

so $\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}$ and $\text{Cov}(X_i, X_j) = \Sigma_{ij}$.

Proof. Write $\mathbf{X} = \boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z}$ with $\mathbf{Z}$ a vector of n independent standard normals, which is (112) after the linear change of variables. With $\mathbf{a} = \Sigma^{1/2} \mathbf{t}$, independence (Theorem 6.7) and the standard normal mgf (56) give $M_{\mathbf{X}}(\mathbf{t}) = e^{\mathbf{t}^\top \boldsymbol{\mu}} \prod_i e^{a_i^2/2} = e^{\mathbf{t}^\top \boldsymbol{\mu} + \mathbf{t}^\top \Sigma \mathbf{t}/2}$. $\square$

Theorem B.10 (Properties; proof omitted). *Let $\mathbf{X} \sim N_n(\boldsymbol{\mu}, \Sigma)$.*

- (a) Affine maps stay normal. *For any $m \times n$ matrix $\mathbf{A}$ of full row rank and any $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{A}\mathbf{X} + \mathbf{b} \sim N_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\Sigma\mathbf{A}^\top)$. In particular every linear combination $\mathbf{a}^\top \mathbf{X}$ is univariate normal, and this characterizes the family.*
- (b) Marginals are normal. *Any subvector of $\mathbf{X}$ is multivariate normal, with the corresponding subvector of $\boldsymbol{\mu}$ and submatrix of Σ .*
- (c) Zero covariance is independence. *Two subvectors are independent if and only if the block of Σ linking them is zero.*
- (d) Conditionals are normal. *Partitioning $\mathbf{X} = (\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top$ conformably with $\boldsymbol{\mu}$ and $\Sigma = (\Sigma_{ij})_{i,j=1,2}$,*

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim N(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}). \quad (114)$$

Parts (a) and (b) follow from (113), (c) from the separation of the quadratic form, and (d) by applying (c) to $\mathbf{W} = \mathbf{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2$, which is uncorrelated with $\mathbf{X}_2$. Part (d) is linear regression in one line: the conditional mean is affine with coefficient matrix $\Sigma_{12}\Sigma_{22}^{-1} = \text{Cov}(\mathbf{X}_1, \mathbf{X}_2) \text{Var}(\mathbf{X}_2)^{-1}$, the population least-squares coefficient, and the conditional variance does not depend on $\mathbf{x}_2$, so a normal population is homoskedastic by construction. Everything the second half of this course does with $\mathbb{E}[\mathbf{Y} \mid \mathbf{X}]$ is an attempt to keep this structure without assuming normality.

C Interchanging Limits with Integrals and Sums

Reading: Casella and Berger [2024, §2.4]. All of the results below follow from Theorem C.1 by applying it along an arbitrary sequence $\delta_n \rightarrow 0$; see Billingsley [1995, Thms. 16.4, 16.8].

Theorem C.1 (Dominated convergence). *Let $h_1, h_2, \dots$ be measurable functions with $h_n(x) \rightarrow h(x)$ for every x , and suppose there is a g with $|h_n(x)| \leq g(x)$ for all n and all x and $\int_{-\infty}^{\infty} g(x) dx < \infty$. Then h is integrable and $\lim_n \int_{-\infty}^{\infty} h_n(x) dx = \int_{-\infty}^{\infty} h(x) dx$.*

Theorem C.2 (Limit and integral). *Suppose $h(x, y)$ is continuous in y at y_0 for each x , and there is a g with $|h(x, y)| \leq g(x)$ for all x and y and $\int_{-\infty}^{\infty} g(x) dx < \infty$. Then*

$$\lim_{y \rightarrow y_0} \int_{-\infty}^{\infty} h(x, y) dx = \int_{-\infty}^{\infty} \lim_{y \rightarrow y_0} h(x, y) dx.$$

C.1 Differentiating under the integral sign

Theorem C.3 (Differentiation under the integral). *Suppose $f(x, \theta)$ is differentiable at θ_0 for every x , and there are a function $g(x, \theta_0)$ and a constant $\delta_0 > 0$ such that $|[f(x, \theta_0 + \delta) - f(x, \theta_0)]/\delta| \leq g(x, \theta_0)$ for all x and all $0 < |\delta| \leq \delta_0$, with $\int_{-\infty}^{\infty} g(x, \theta_0) dx < \infty$. Then*

$$\frac{d}{d\theta} \int_{-\infty}^{\infty} f(x, \theta) dx \Big|_{\theta=\theta_0} = \int_{-\infty}^{\infty} \frac{\partial}{\partial \theta} f(x, \theta) \Big|_{\theta=\theta_0} dx. \quad (115)$$

Proof. Let $\delta_n \rightarrow 0$ with $0 < |\delta_n| \leq \delta_0$ and put $h_n(x) = [f(x, \theta_0 + \delta_n) - f(x, \theta_0)]/\delta_n$. Differentiability gives $h_n(x) \rightarrow \partial_{\theta} f(x, \theta_0)$ for every x , and the hypothesis dominates h_n by an integrable g , so Theorem C.1 applies to the difference quotients of $\theta \mapsto \int f(x, \theta) dx$. The sequence being arbitrary, the derivative exists and equals the right-hand side. $\square$

Corollary C.4. *The conclusion (115) holds at every θ if $f(x, \cdot)$ is differentiable on a neighbourhood of θ and there are $g(x, \theta)$ and $\delta_0 > 0$ with*

$$\left| \frac{\partial}{\partial \theta} f(x, \theta) \Big|_{\theta=\theta'} \right| \leq g(x, \theta) \quad \text{for all } \theta' \text{ with } |\theta' - \theta| \leq \delta_0, \quad (116)$$

and $\int_{-\infty}^{\infty} g(x, \theta) dx < \infty$.

Proof. By the mean value theorem the difference quotient in Theorem C.3 equals $\partial_\theta f(x, \theta')$ for some θ' within δ_0 of θ , which (116) dominates by $g(\cdot, \theta)$. $\square$

C.2 Sums

Theorem C.5 (Differentiation under a sum; proof omitted). *Suppose $\sum_{x=0}^{\infty} h(\theta, x)$ converges for every θ in an interval (a, b) , that $\frac{\partial}{\partial \theta} h(\theta, x)$ is continuous in θ for each x , and that $\sum_{x=0}^{\infty} \frac{\partial}{\partial \theta} h(\theta, x)$ converges uniformly on every closed bounded subinterval of (a, b) . Then*

$$\frac{d}{d\theta} \sum_{x=0}^{\infty} h(\theta, x) = \sum_{x=0}^{\infty} \frac{\partial}{\partial \theta} h(\theta, x). \quad (117)$$

Theorem C.6 (Sum and integral; proof omitted). *Suppose $\sum_{x=0}^{\infty} h(\theta, x)$ converges uniformly on $[a, b]$ and that $h(\cdot, x)$ is continuous on $[a, b]$ for each x . Then*

$$\int_a^b \sum_{x=0}^{\infty} h(\theta, x) d\theta = \sum_{x=0}^{\infty} \int_a^b h(\theta, x) d\theta.$$

Both are the counting-measure versions of Theorem C.1: uniform convergence supplies the dominating function.